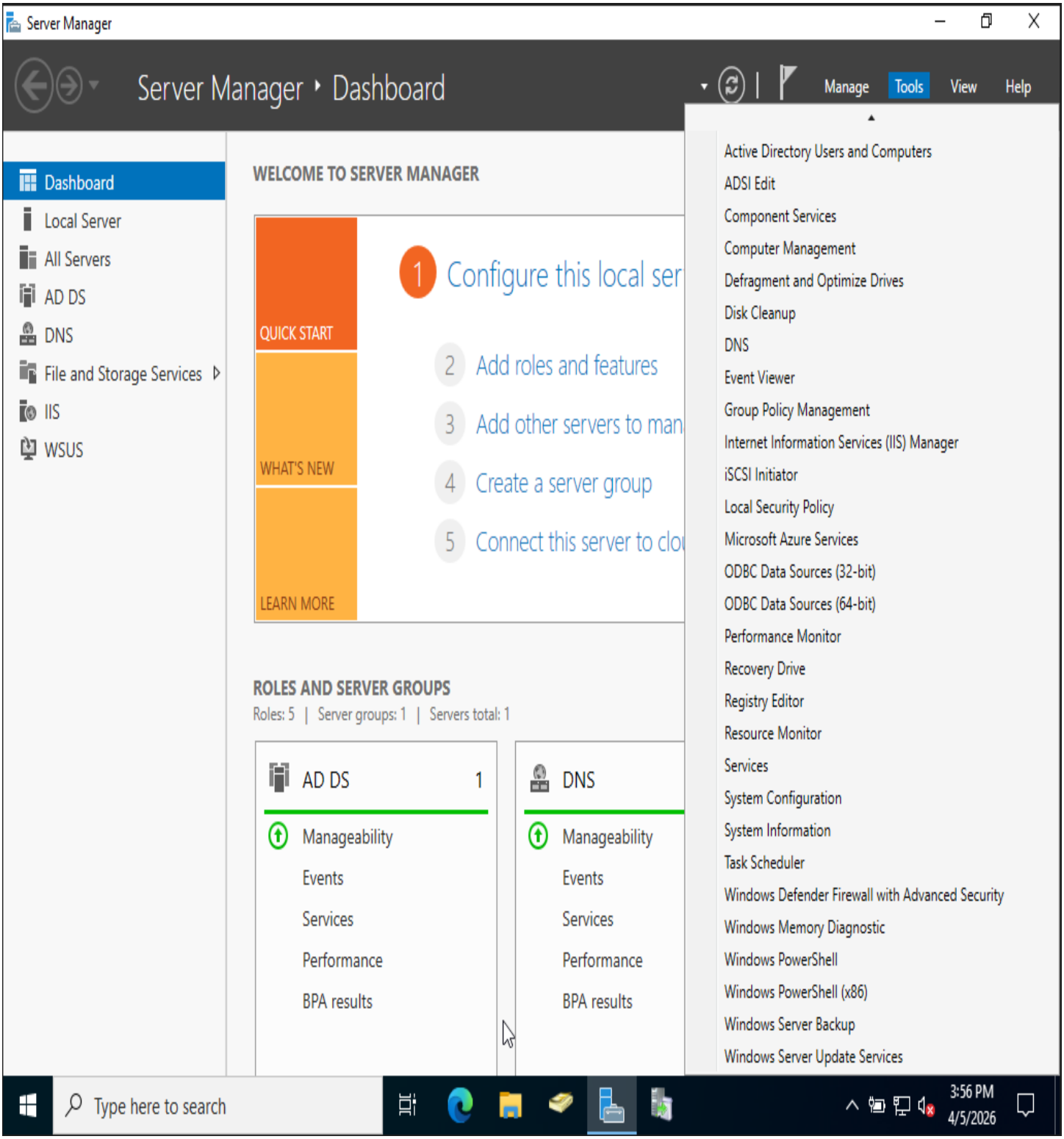

ENTERPRISE IDENTITY & ENDPOINT MANAGEMENT LAB

By Elie Nzweme

Enterprise Identity & Endpoint Management Lab

1. Project Overview

Objective

This project simulates a real-world enterprise IT environment where I designed, deployed, and supported a Windows-based infrastructure.
The goal was to gain hands-on experience in:

- System administration
- Enterprise troubleshooting
- Help desk workflows
- Patch management and endpoint support

Technologies Implemented

Identity & Access

- Active Directory (AD DS)

Endpoint Management

- WSUS, GPO

Support Tools

- Jira Service Management

2. Domain Controller Deployment

Tasks Completed

- Installed Windows Server 2022
- Assigned static IP: **192.168.0.49**
- Installed Active Directory Domain Services (AD DS)
- Promoted server to Domain Controller
- Created domain: **dc.local**
- Configured DNS

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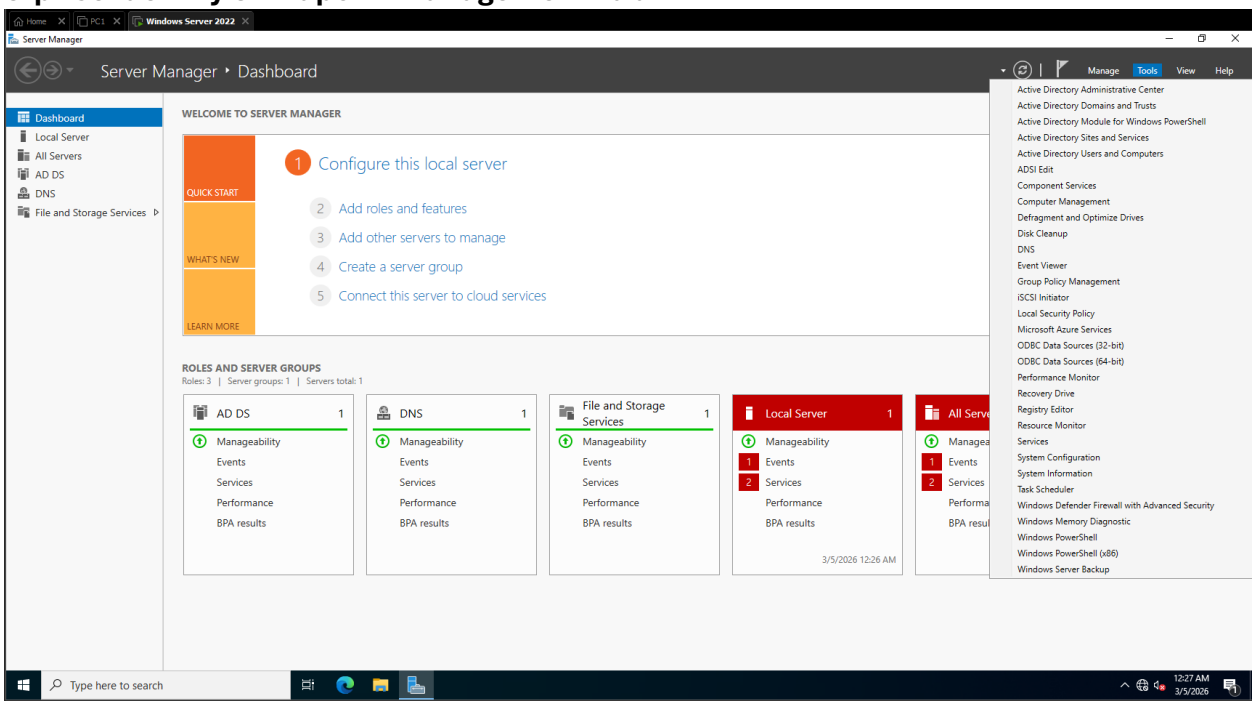

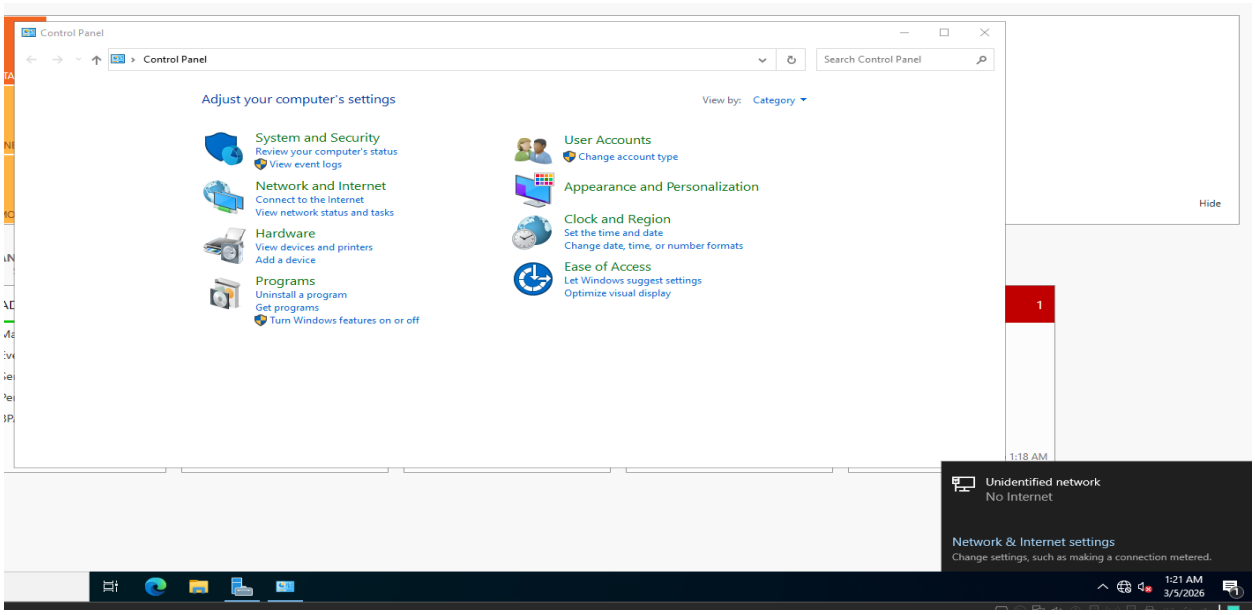

Issues & Troubleshooting

Issue: DNS Misconfiguration

Symptoms:

- Clients unable to resolve domain name
- Domain join failed

Troubleshooting Steps:

- Checked DNS settings on client machines
- Verified DNS service on Domain Controller

Resolution:

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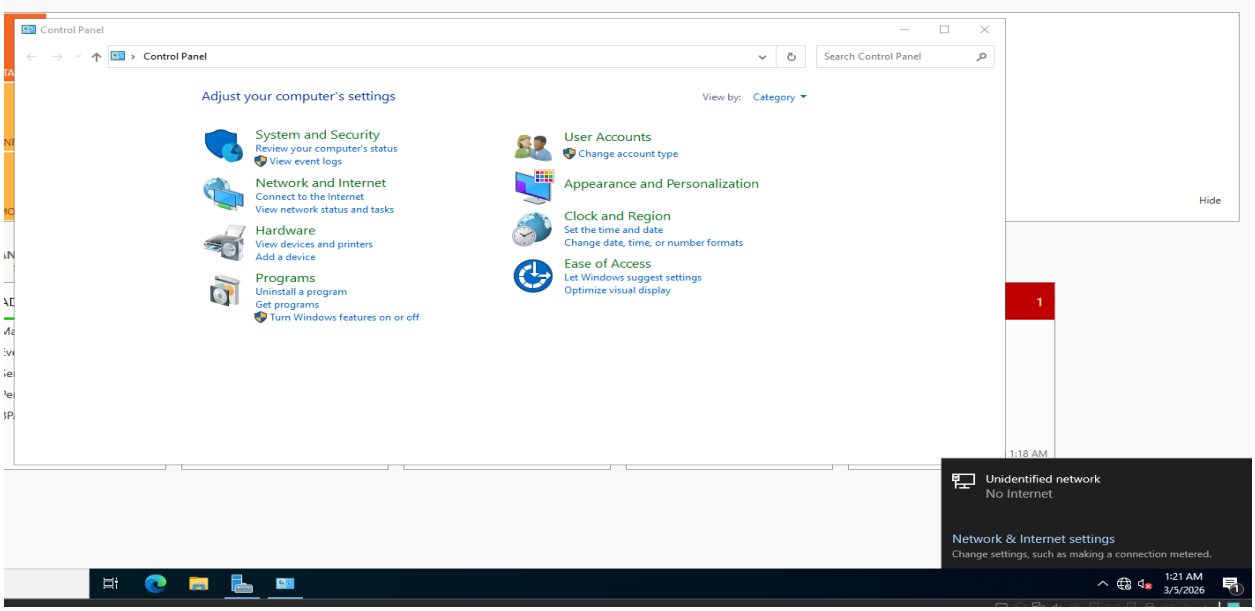
- Configured client DNS to point to Domain Controller (192.168.0.49)
- Confirmed DNS service running

Issue: Network Connectivity Loss

Symptoms:

- Loss of connectivity after static IP configuration



Troubleshooting Steps:

- Reviewed IP configuration
- Tested connectivity using ping

Resolution:

- Corrected gateway and DNS settings
- Restored network connectivity

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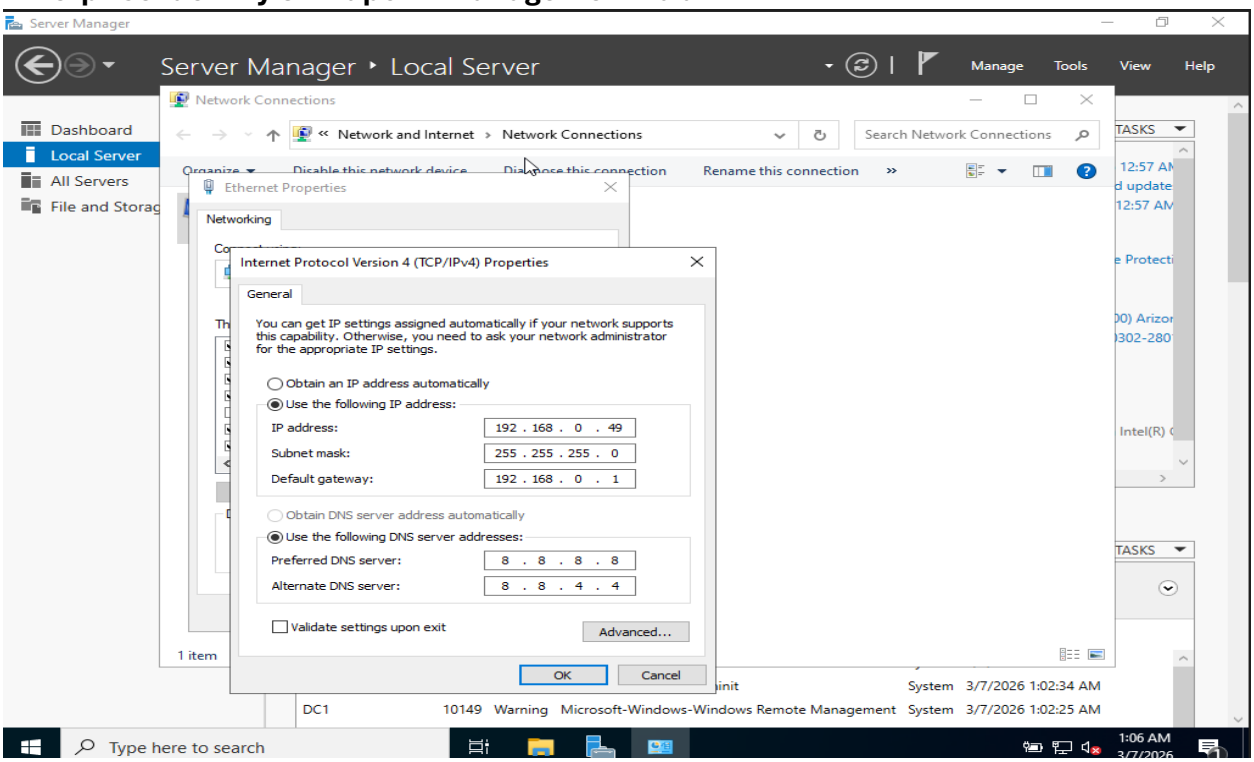
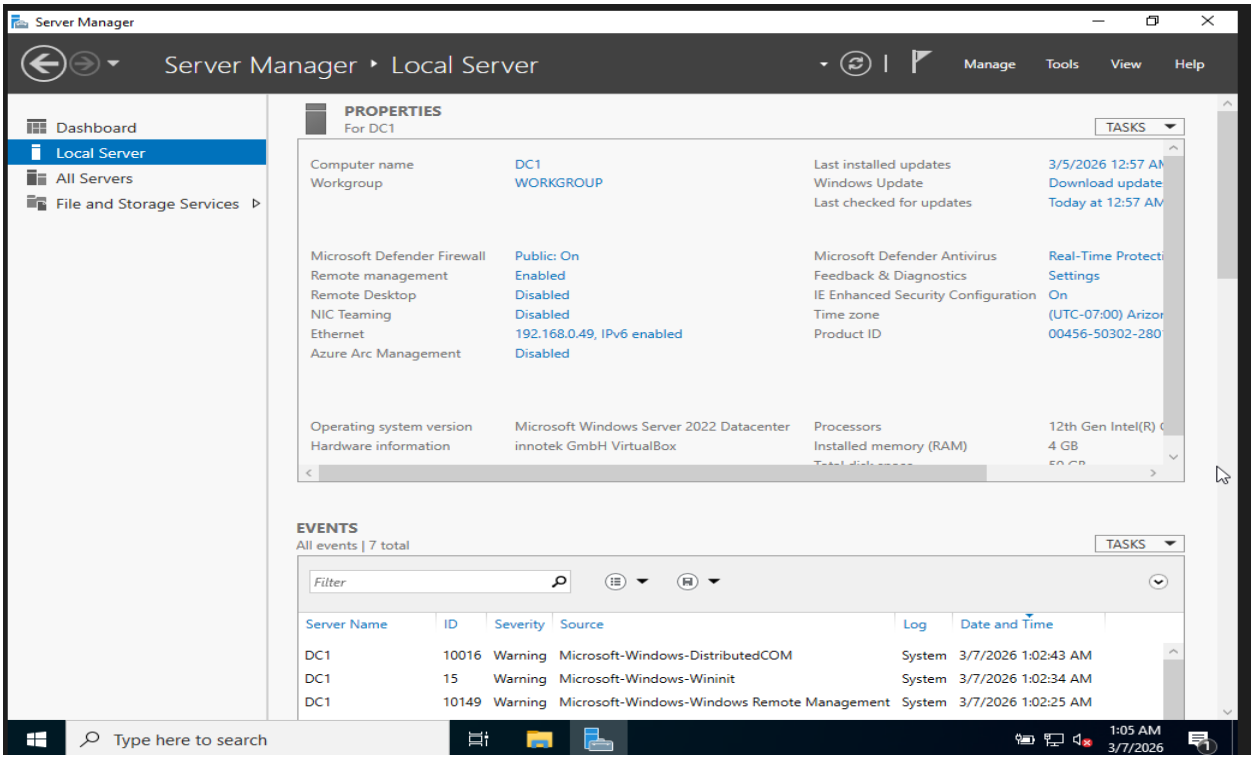
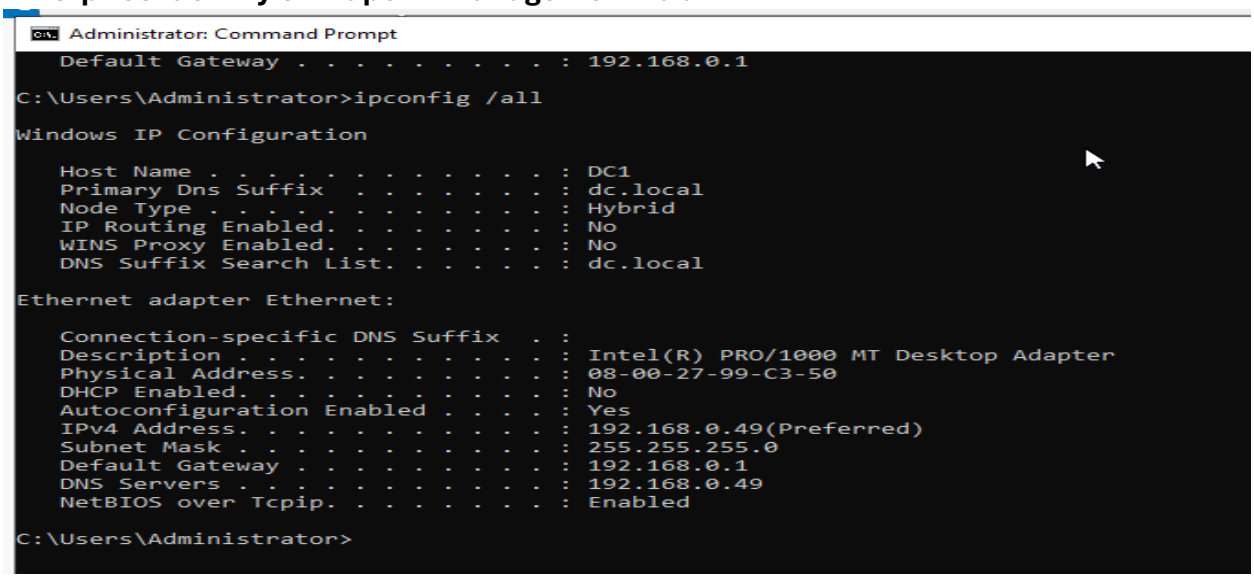






3. Domain Join (Client Machines)

Tasks Completed

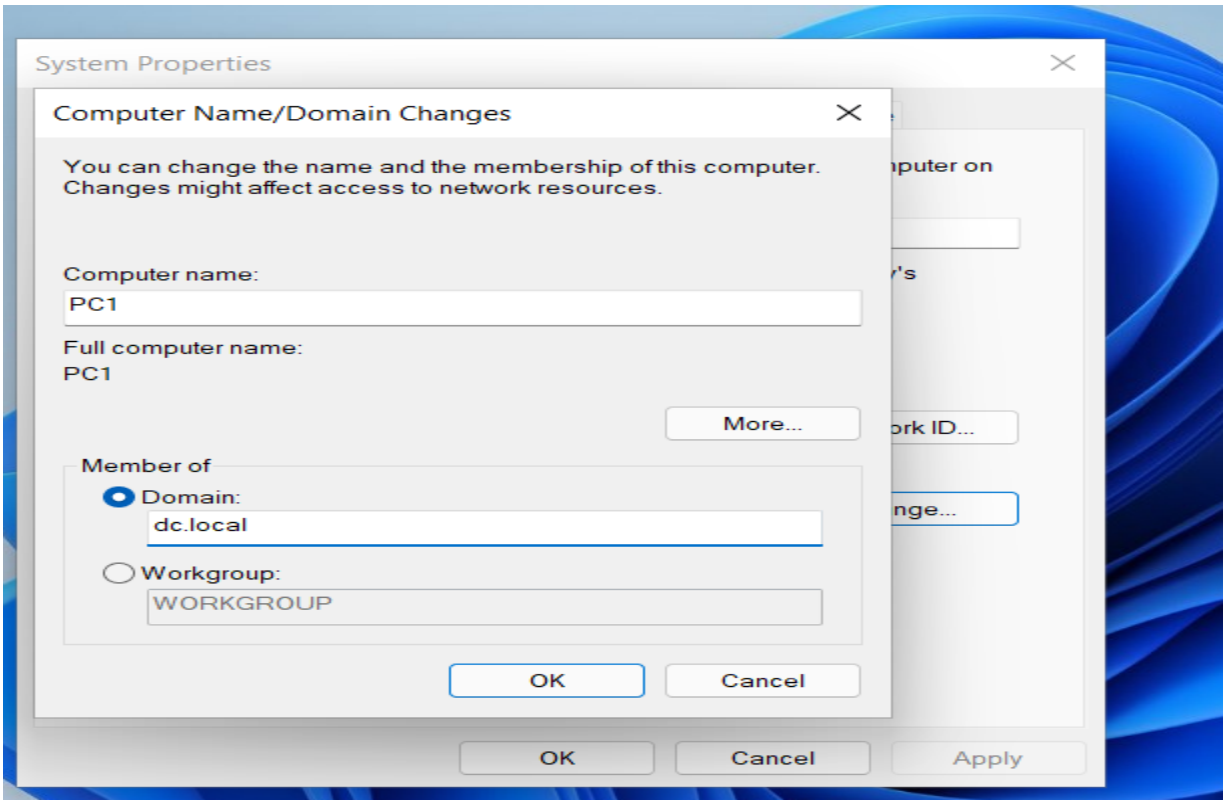
- Configured client DNS
- Joined Windows 10/11 machines to domain
- Authenticated using domain credentials
- Verified successful login

Issues & Troubleshooting

Issue: Domain Join Failure

Symptoms:

- Client unable to locate domain



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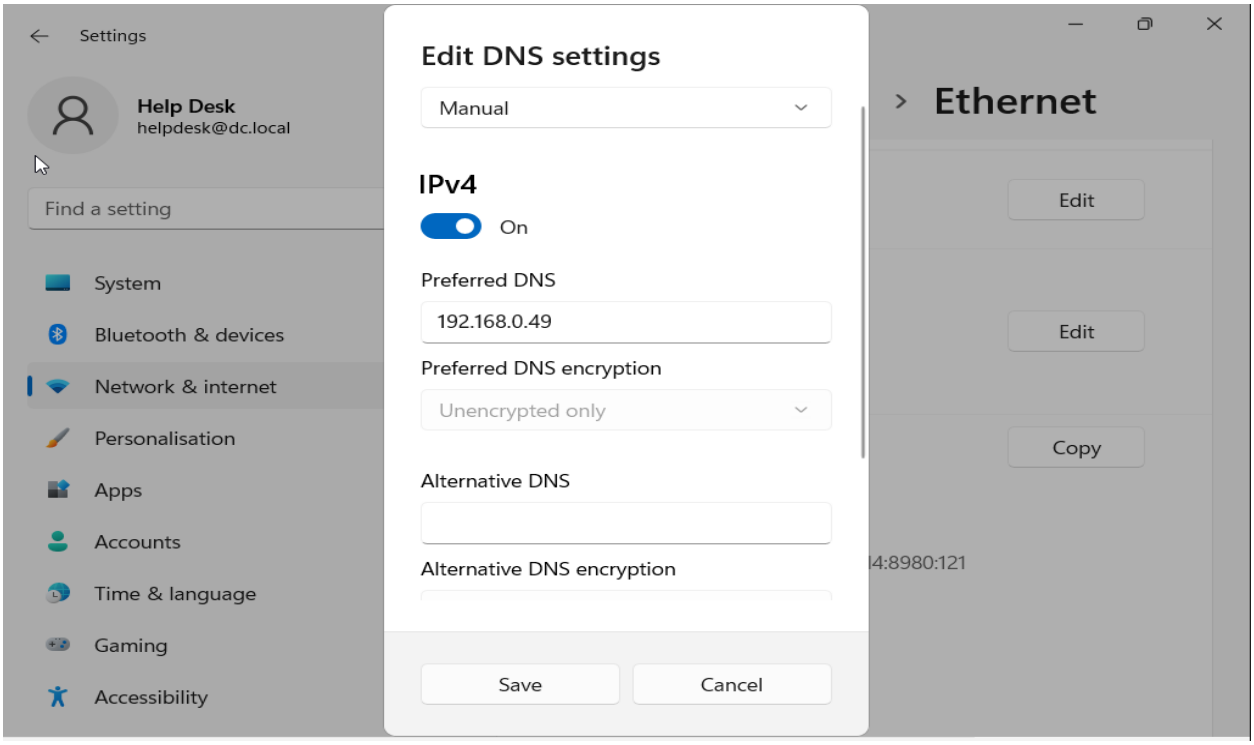
Troubleshooting Steps:

- Verified DNS configuration
- Checked domain name spelling
- Restarted client

Resolution:

- Correct DNS configuration
- Successful domain join



Issue: Login Problems

Symptoms:

- Unable to log in with domain account

Troubleshooting Steps:

- Tested login formats:
  - domain\username
  - .\Administrator
- Verified user account exists in Active Directory

Resolution:

- Used correct login format
- Confirmed account authentication

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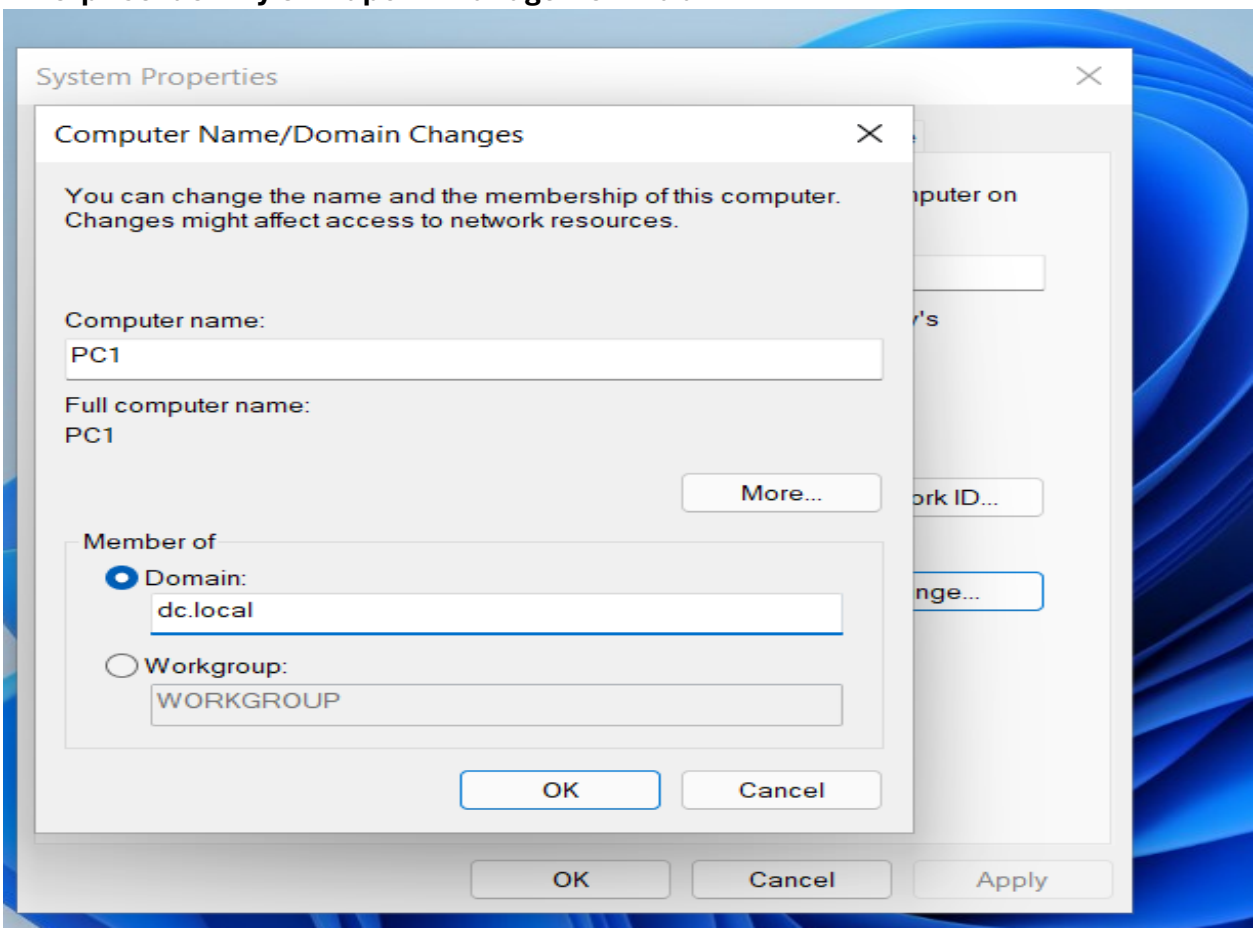

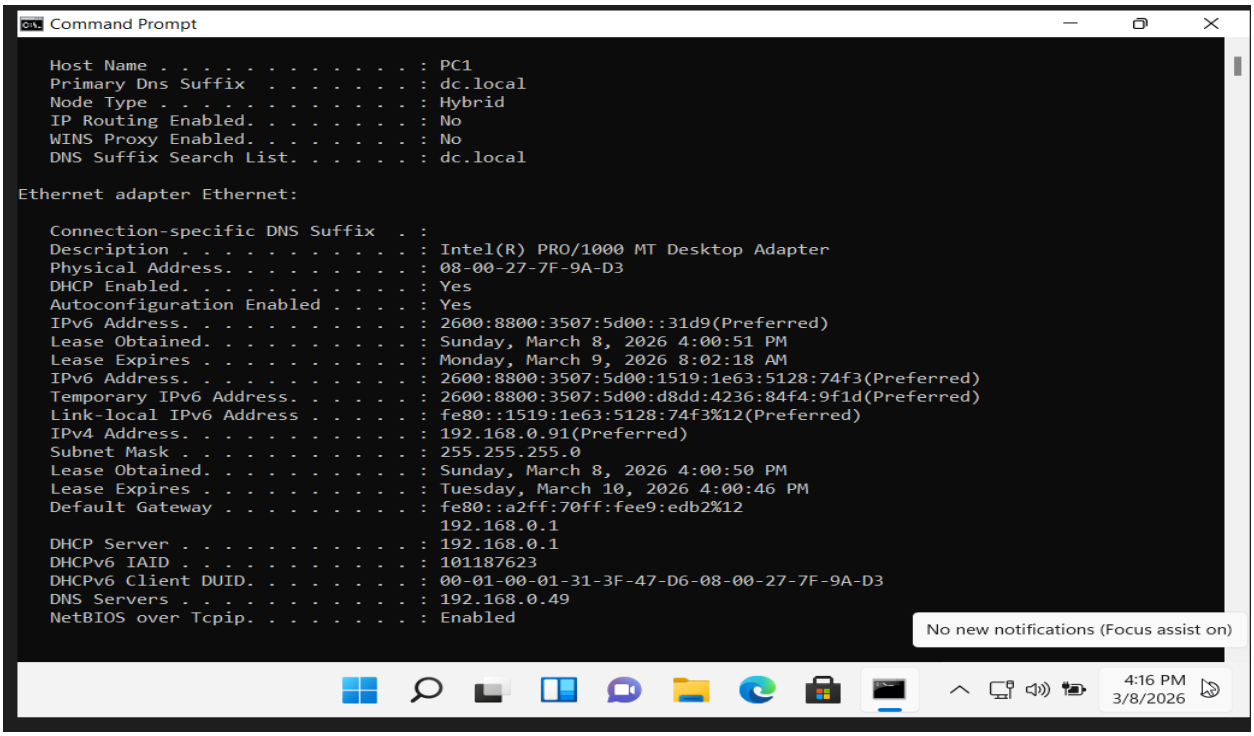

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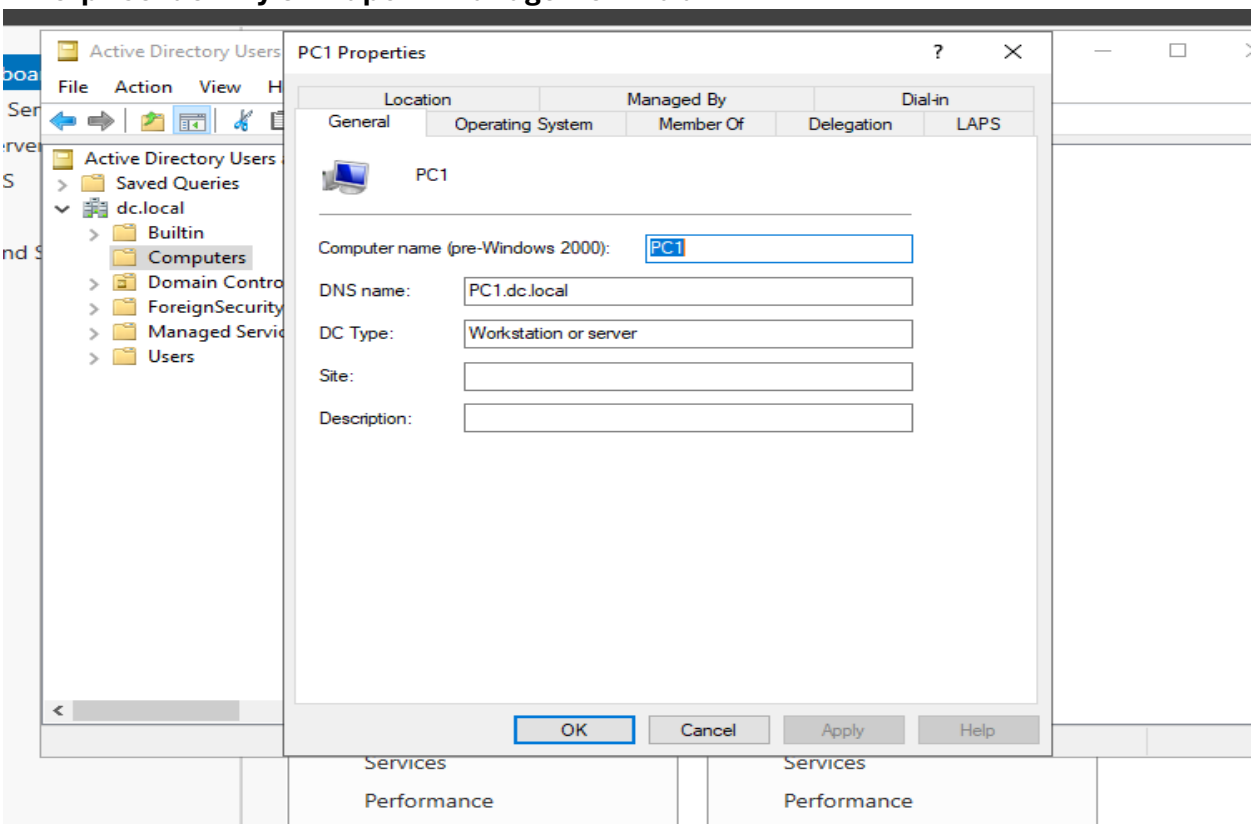

4. Active Directory Configuration

Tasks Completed

- Created Organizational Units (OUs):
  - IT
  - Engineering Team
  - Finance
  - Sales
- Created users and security groups
- Assigned users to appropriate groups

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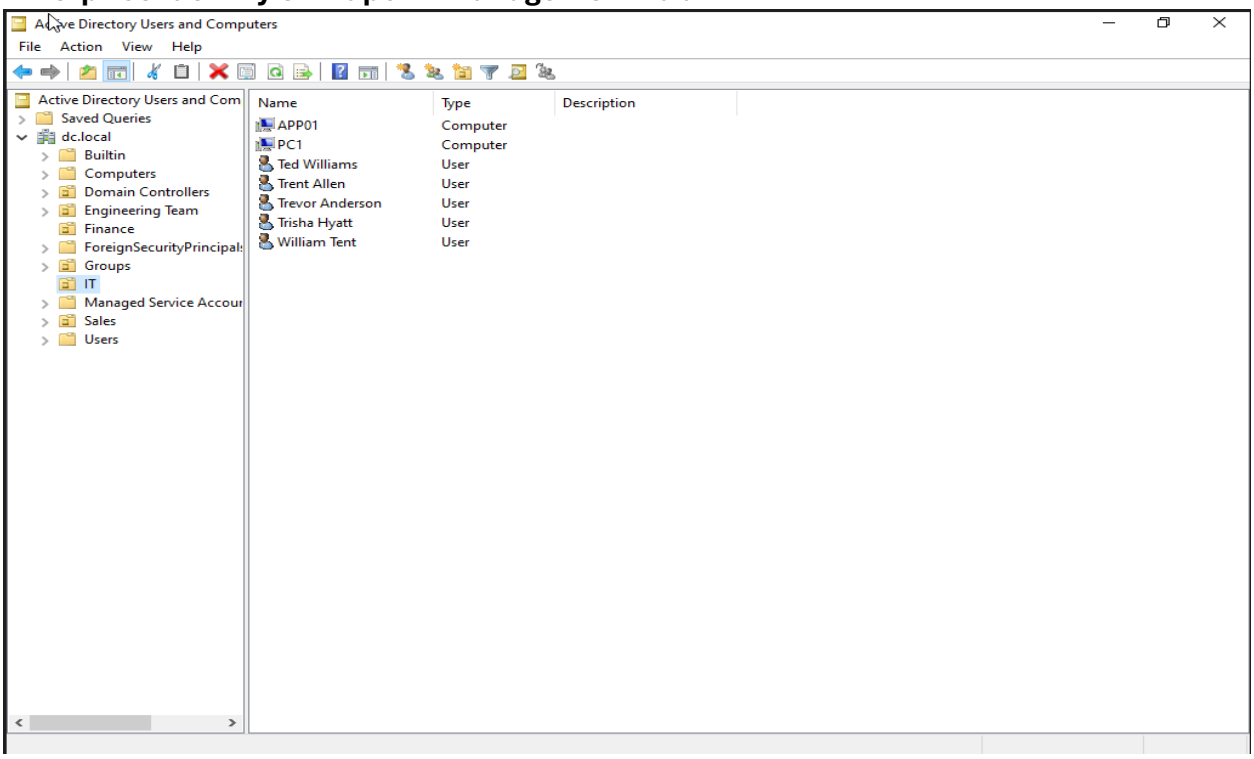

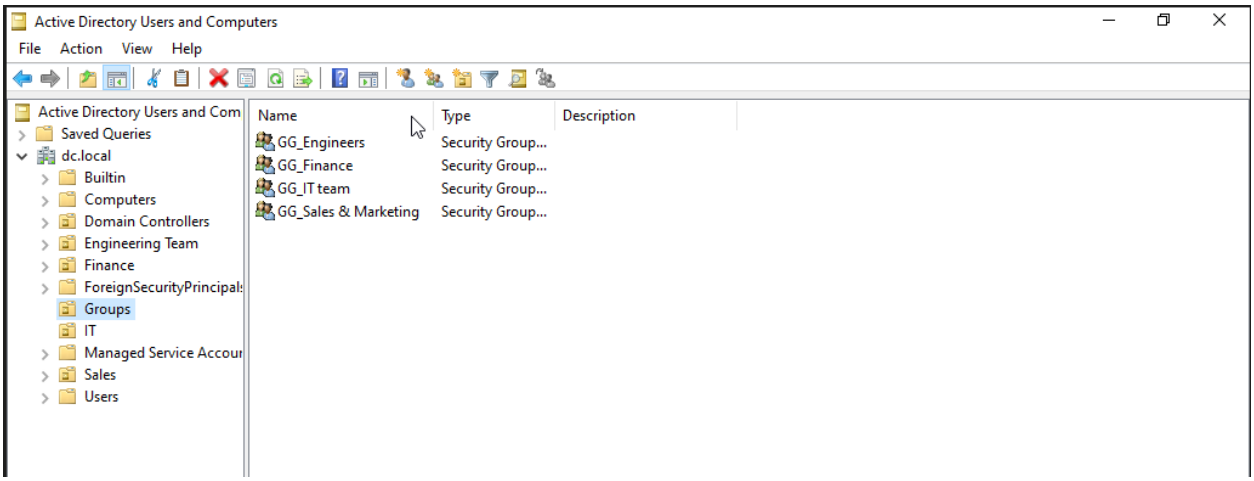

Issues & Troubleshooting

Issue: Incorrect Group Membership

Symptoms:

- Users unable to access resources

Troubleshooting Steps:

- Reviewed group memberships
- Checked permission inheritance

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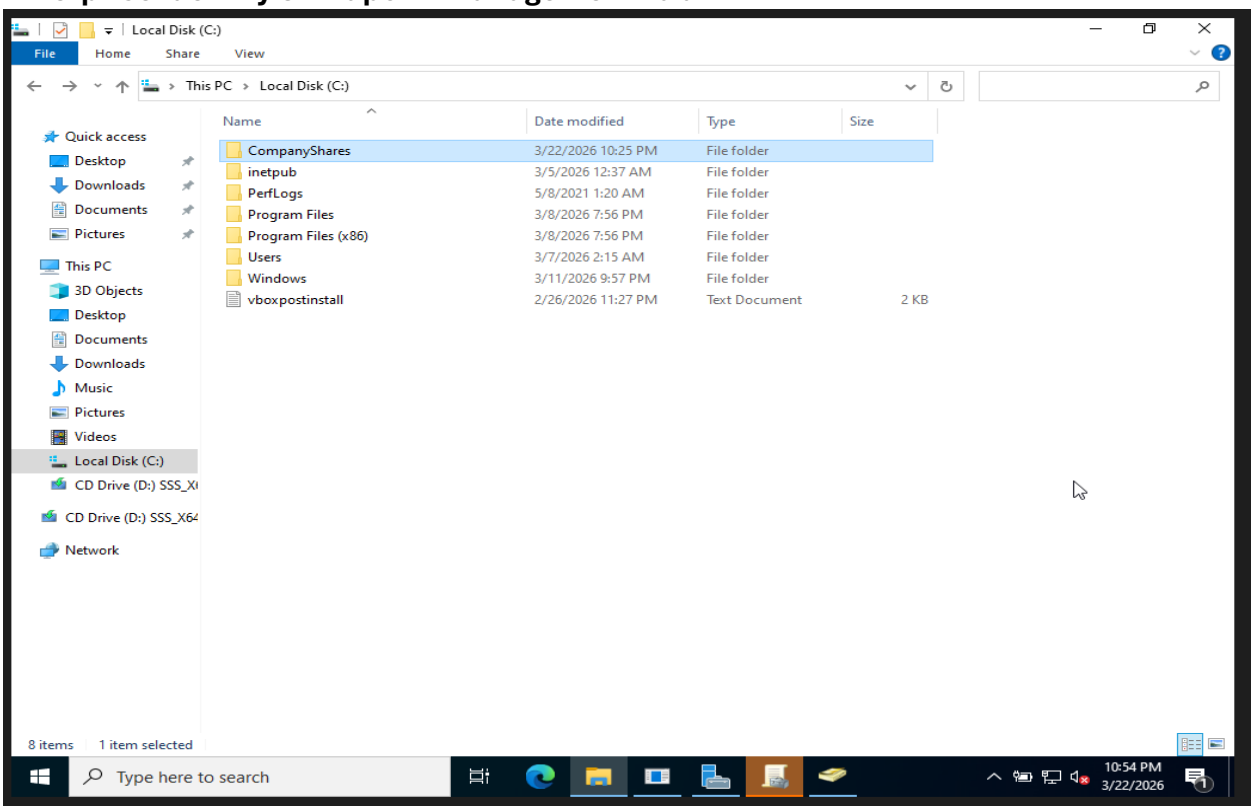

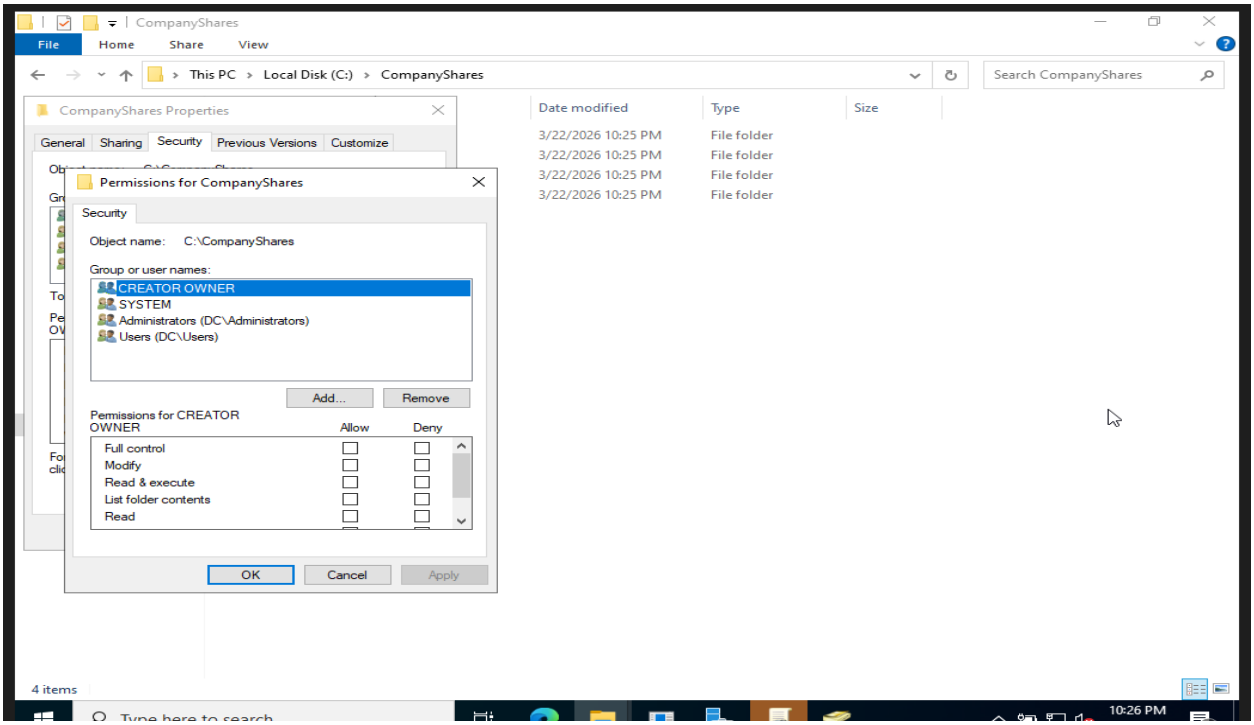

Resolution:

- Updated group assignments
- Verified access

5. File Sharing & Permissions

Tasks Completed

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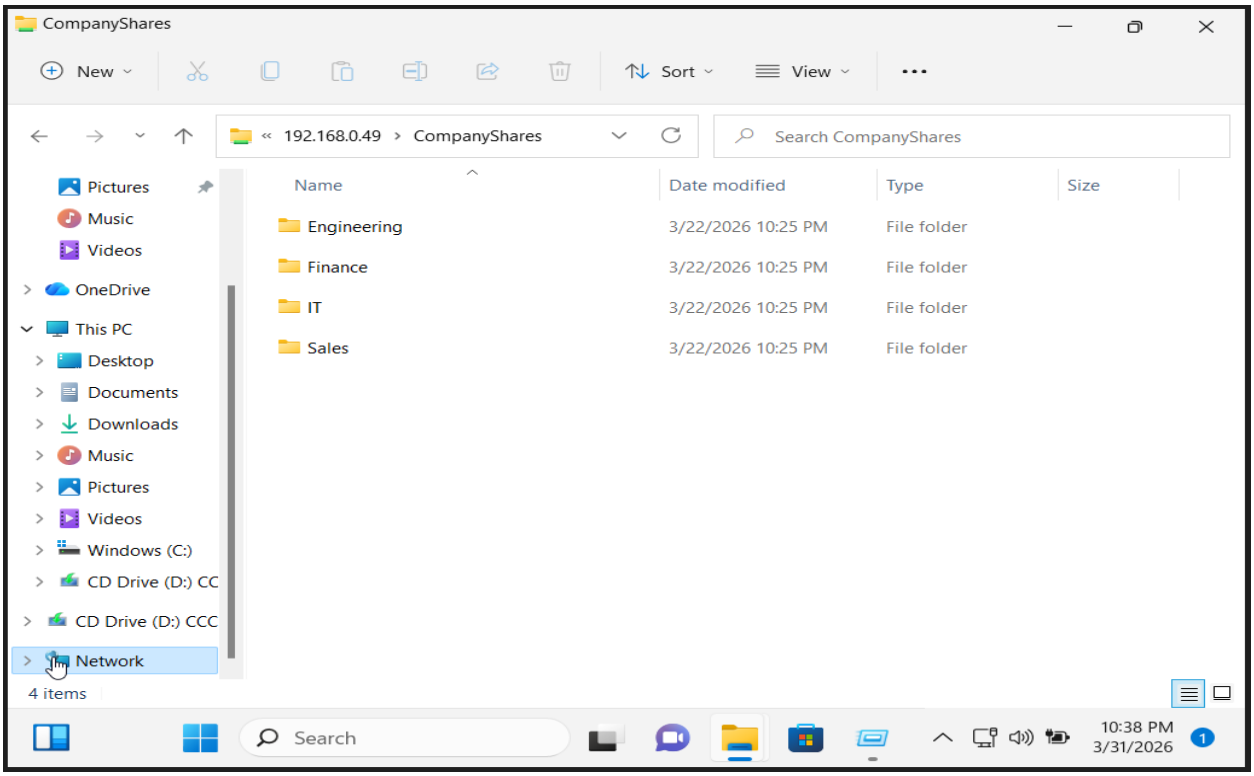
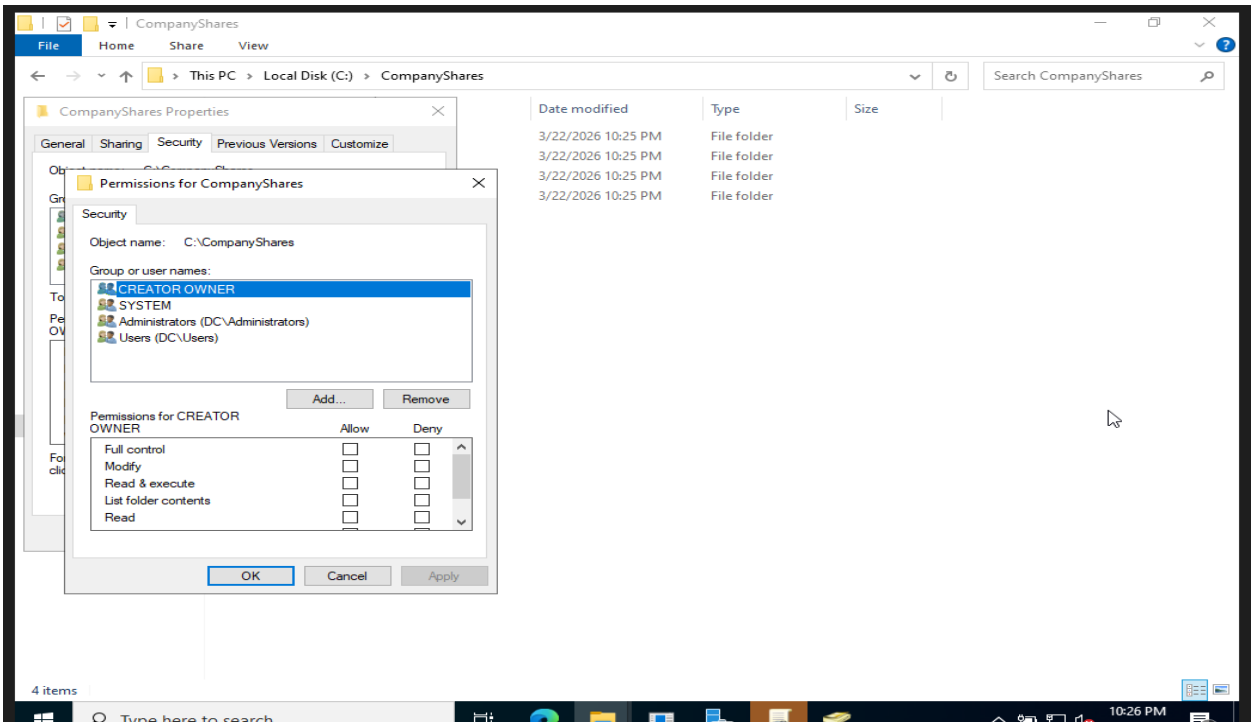
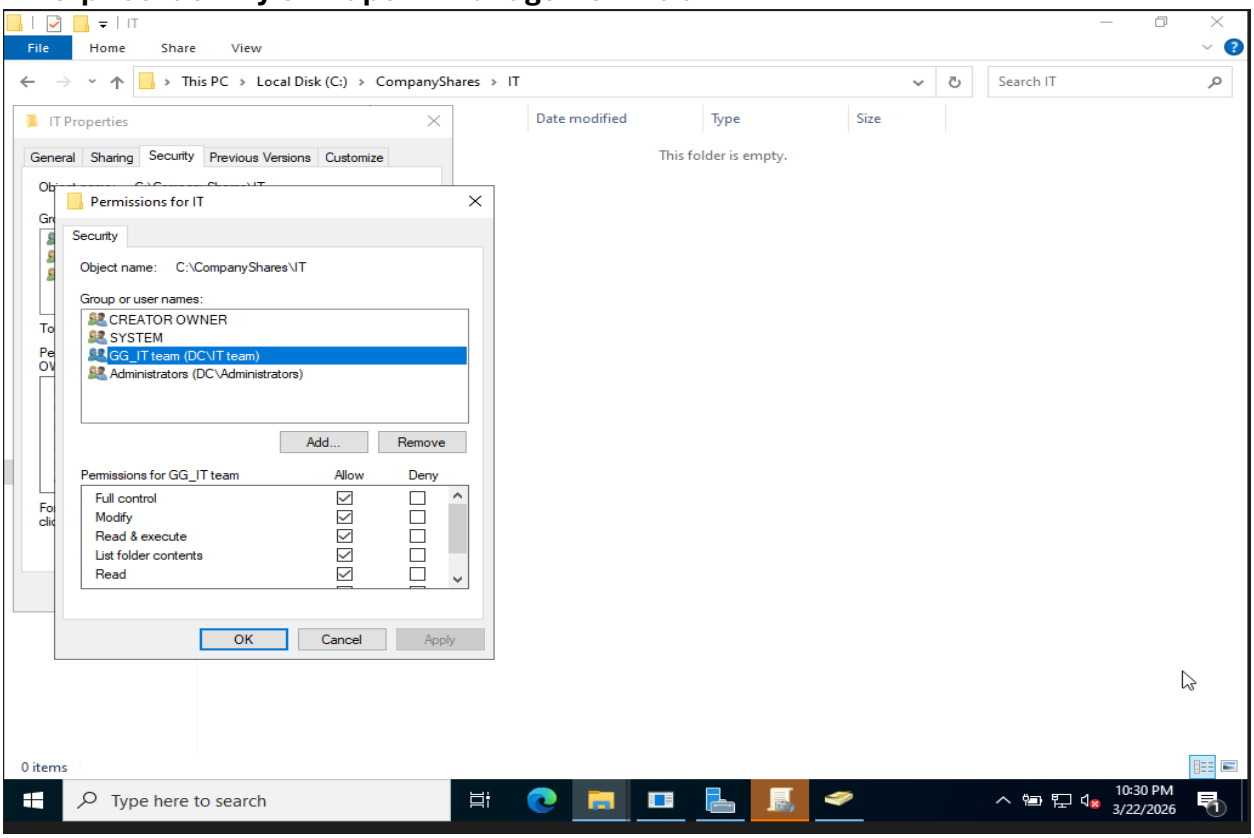
- Created shared folder:

\\192.168.0.49\CompanyShares

- Created departmental folders
- Configured:
  - NTFS permissions
  - Share permissions







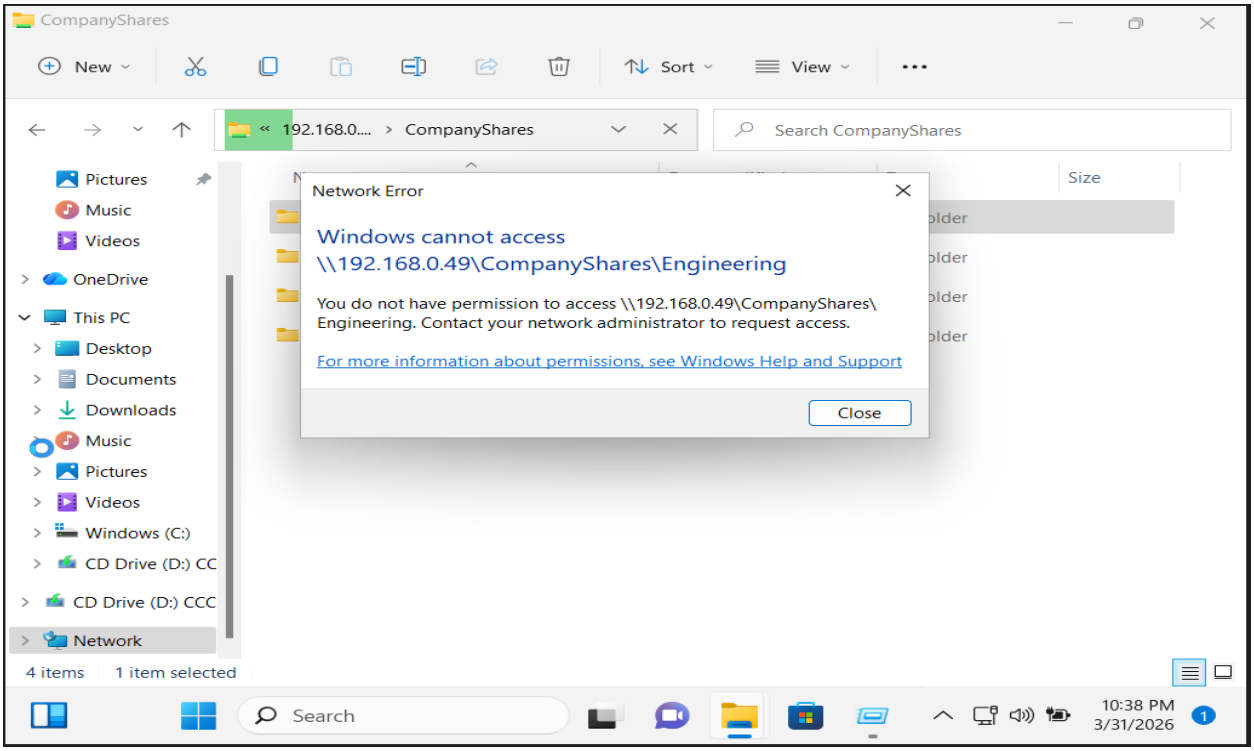
- Enabled Access-Based Enumeration (ABE)

Issues & Troubleshooting

Issue: Unauthorized Folder Access

Symptoms:

- Users could see/access other department folders



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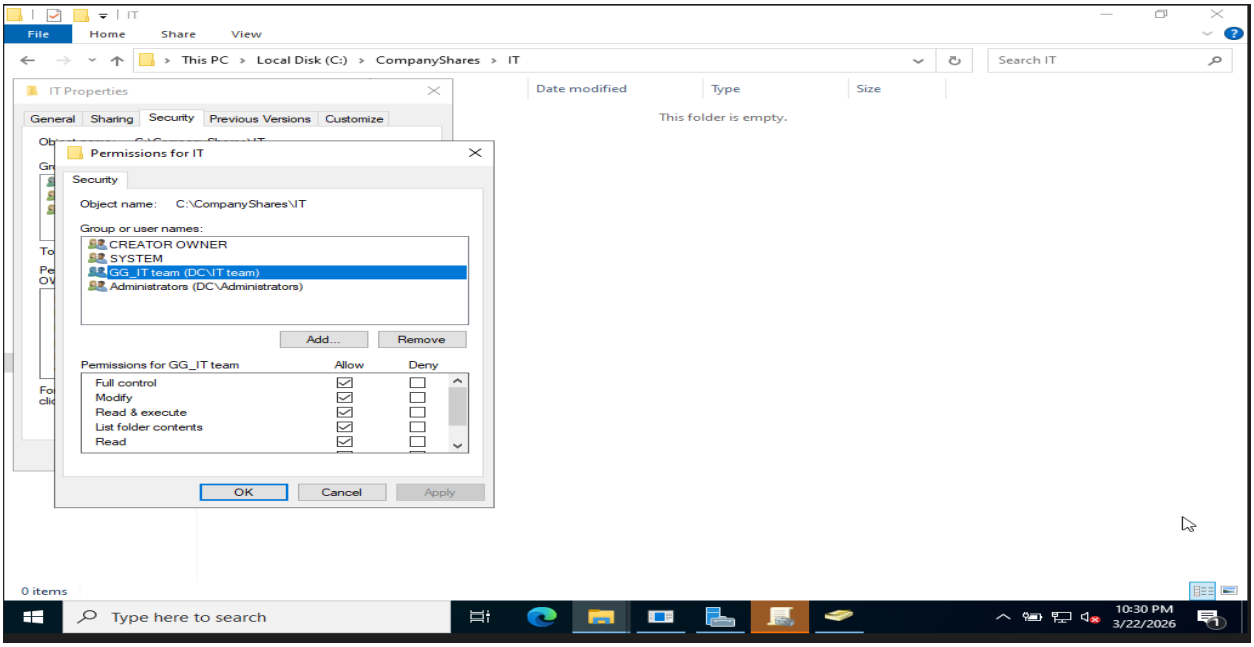
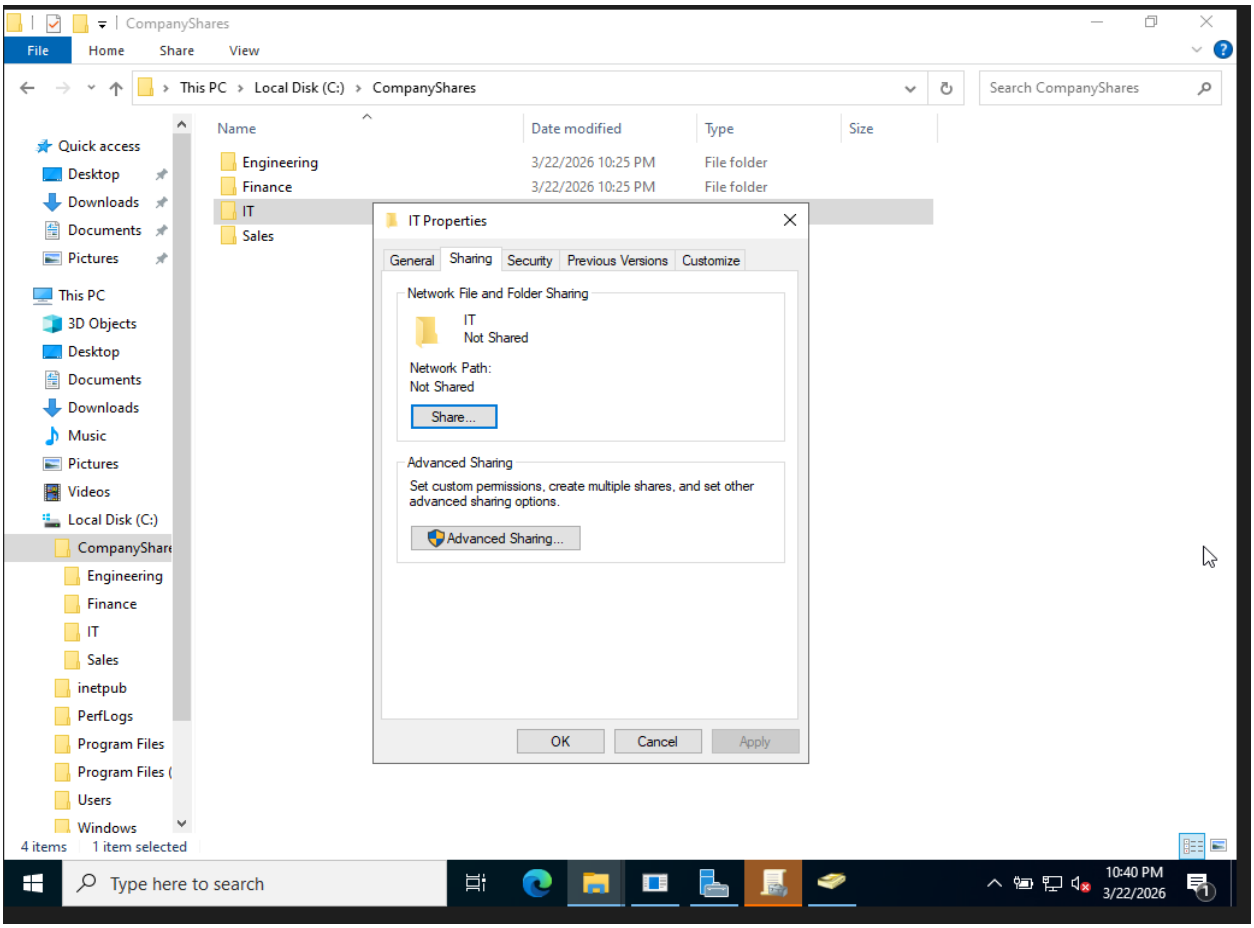
Resolution:

- Corrected NTFS permissions
- Enabled ABE to hide unauthorized folders





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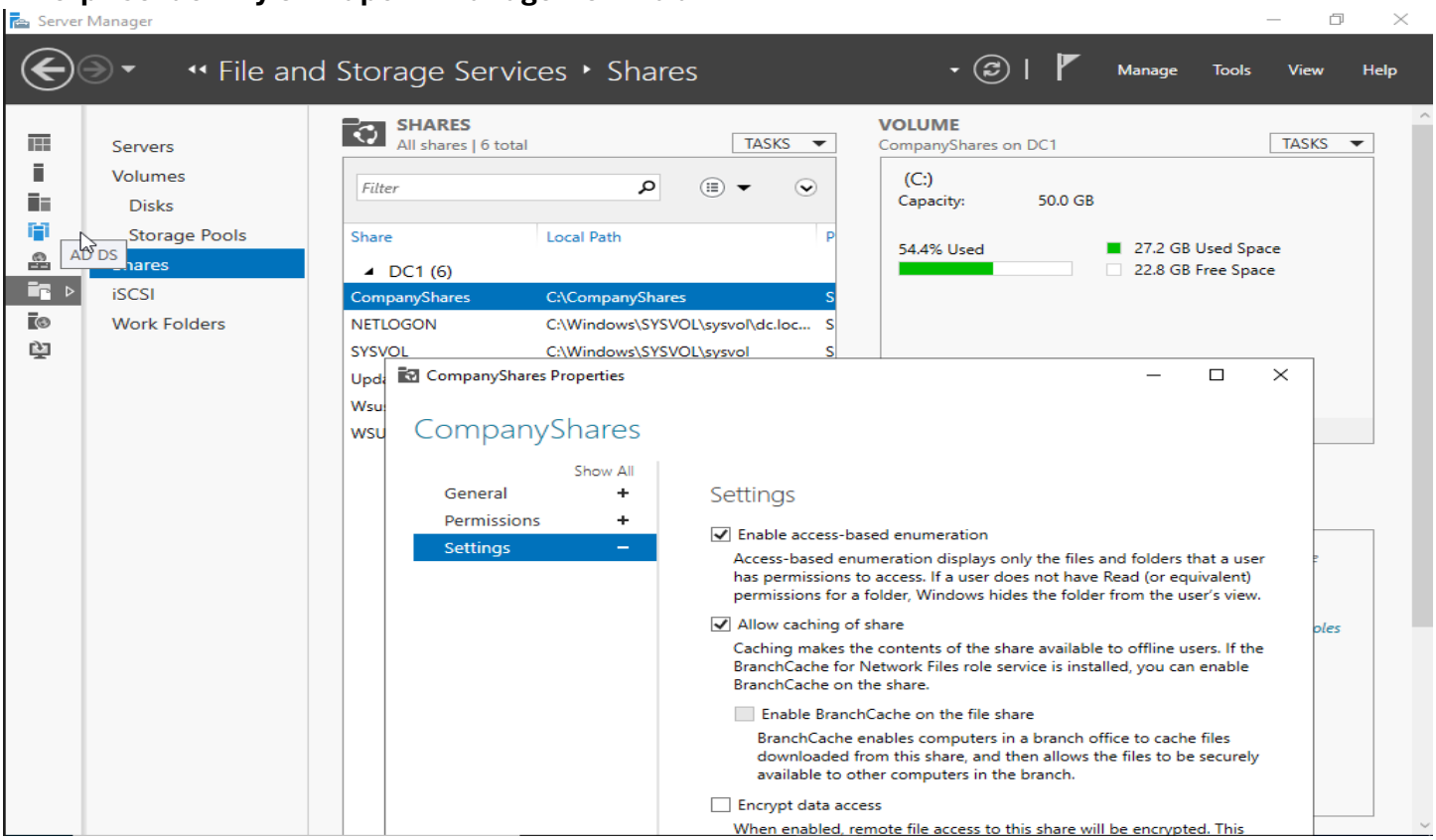

Issue: Access Denied Errors

Symptoms:

- Users unable to access correct folders

Troubleshooting Steps:

- Reviewed security group permissions
- Tested access with different users

Resolution:

- Adjusted permissions
- Confirmed correct access

6. Group Policy (GPO)

Tasks Completed

- Created GPO for drive mapping
- Linked GPO to department OUs
- Configured automatic drive mapping

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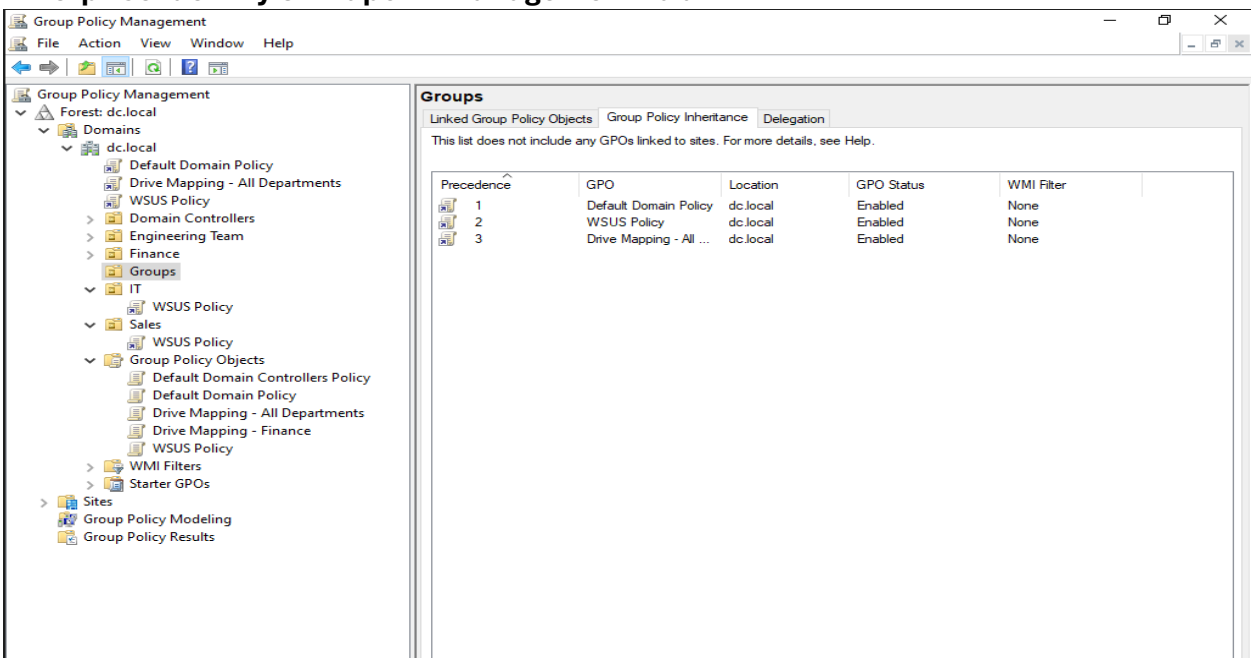
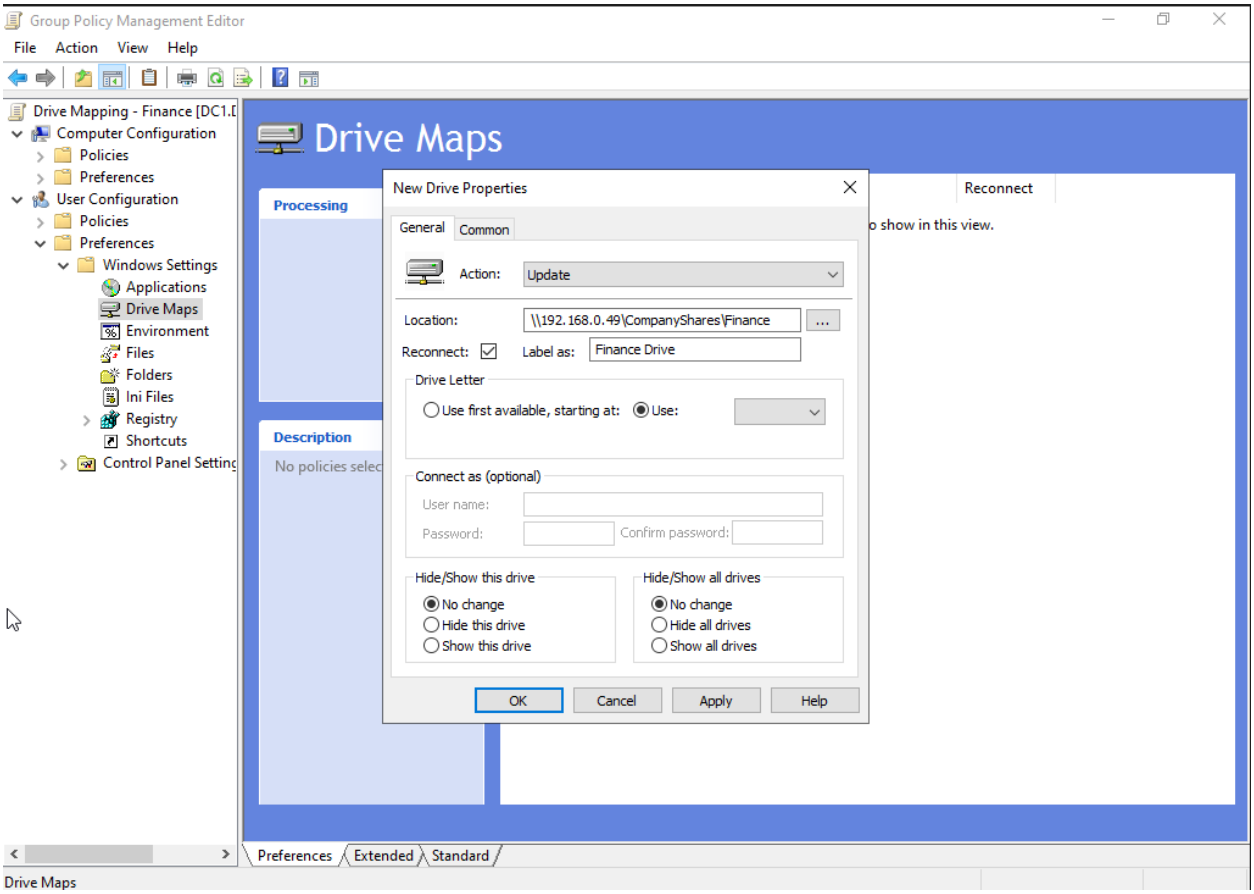
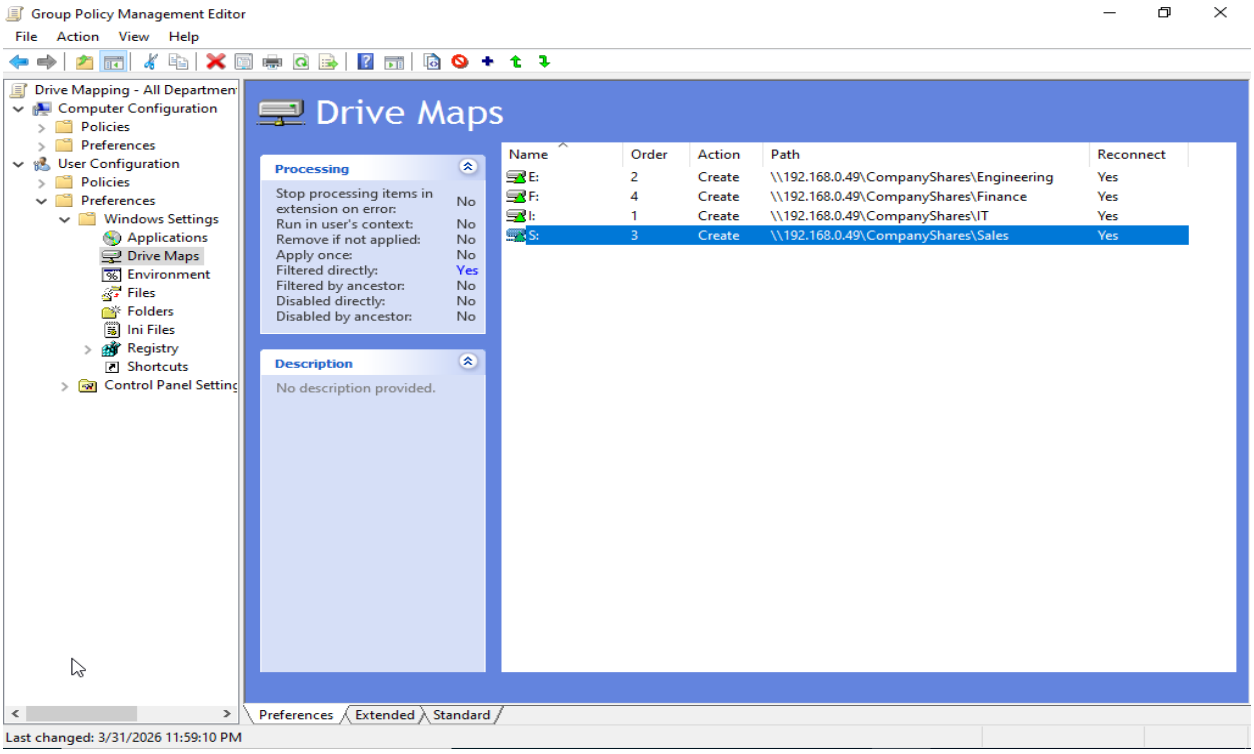




Enterprise Identity & Endpoint Management Lab



Issues & Troubleshooting

Issue: GPO Not Applying

Symptoms:

- Drives not mapping

Troubleshooting Steps:

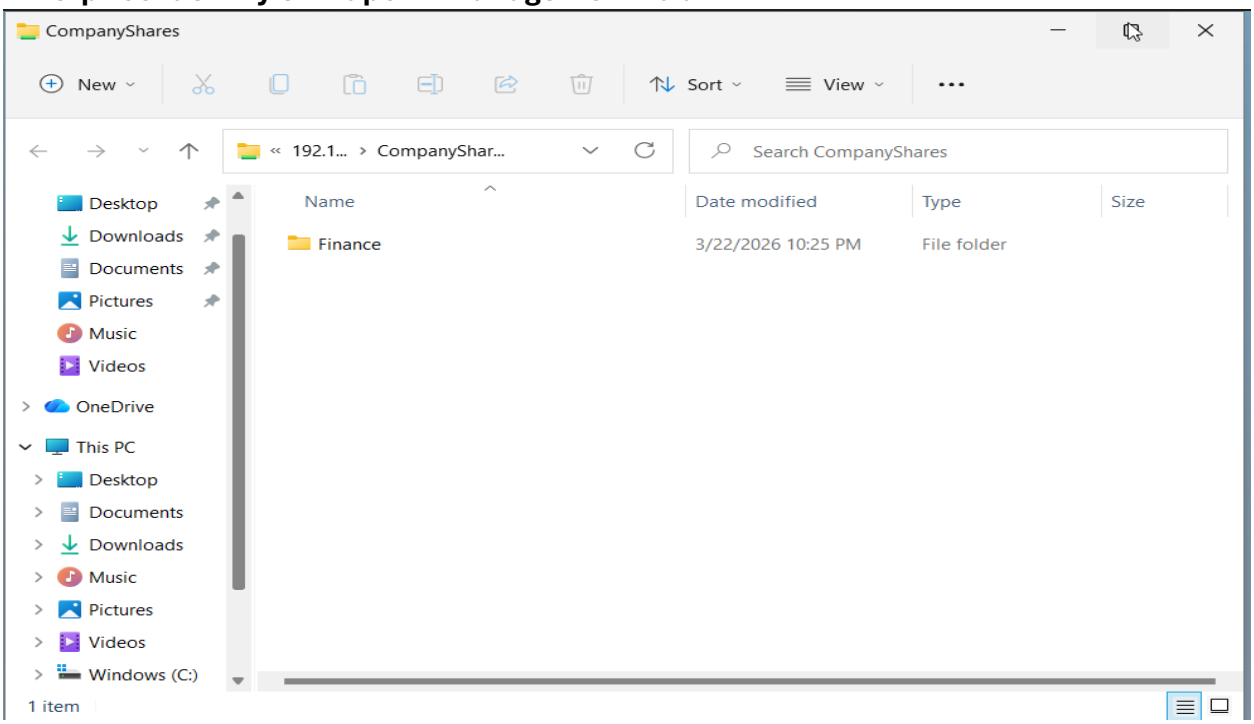
- Ran:

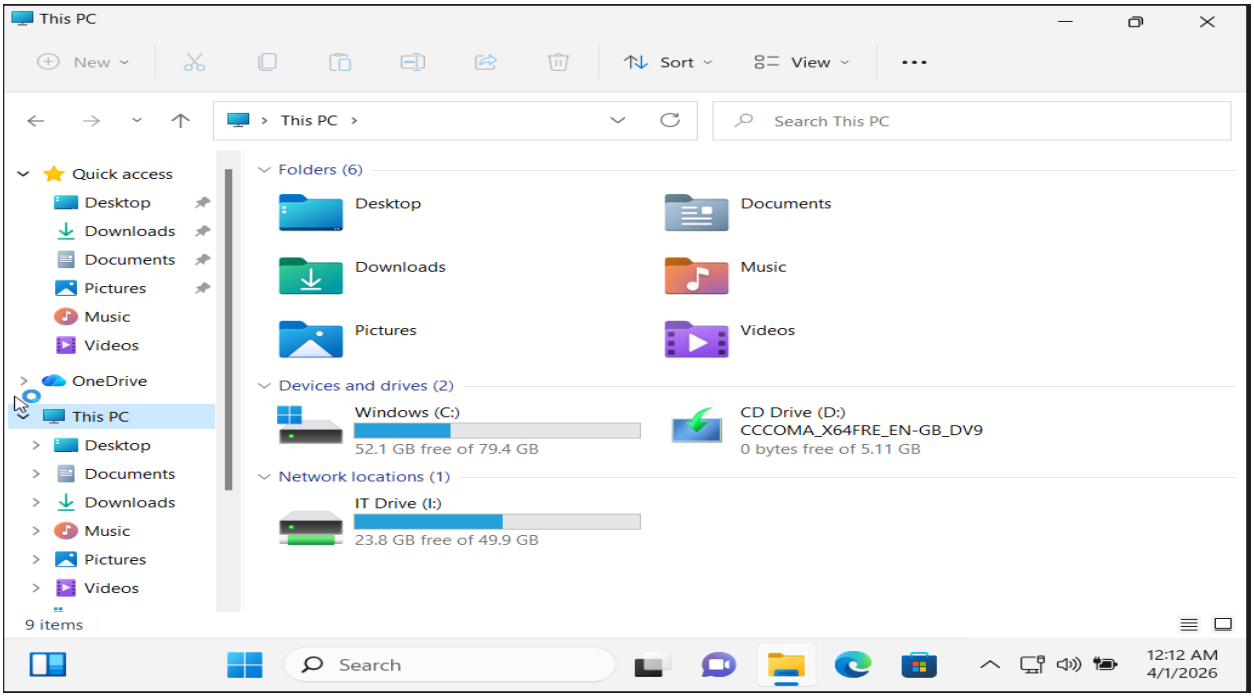
gpupdate /force

- Restarted client
- Verified OU placement

Resolution:

- GPO successfully applied
- Drives mapped correctly





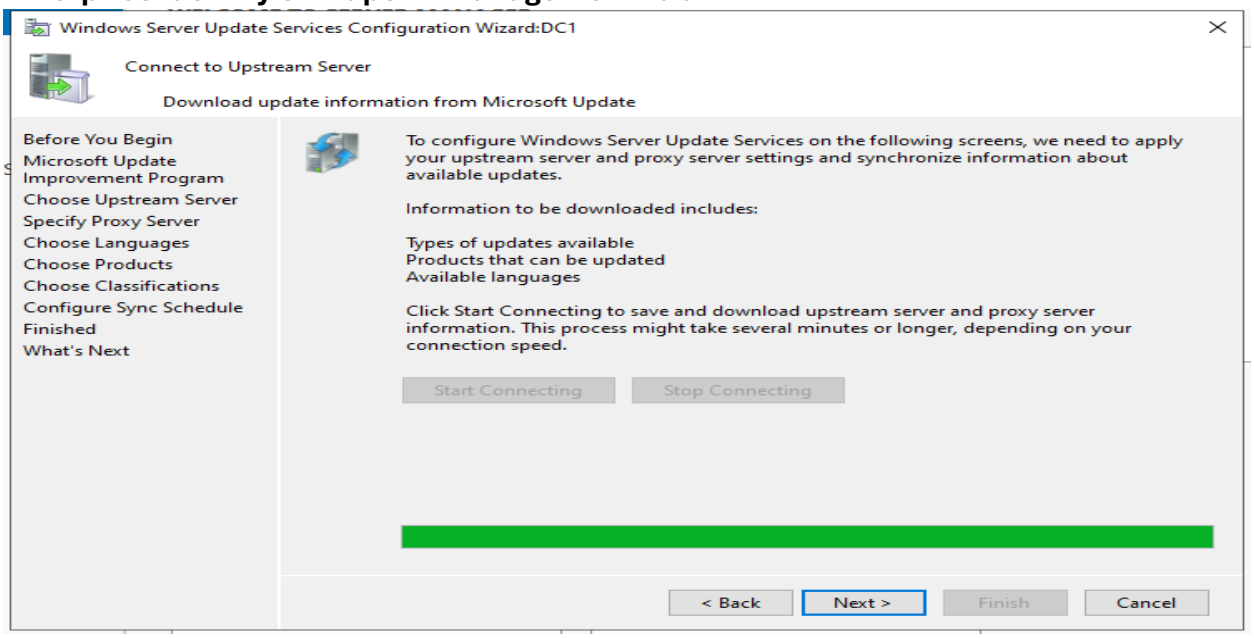
7. WSUS (Patch Management)

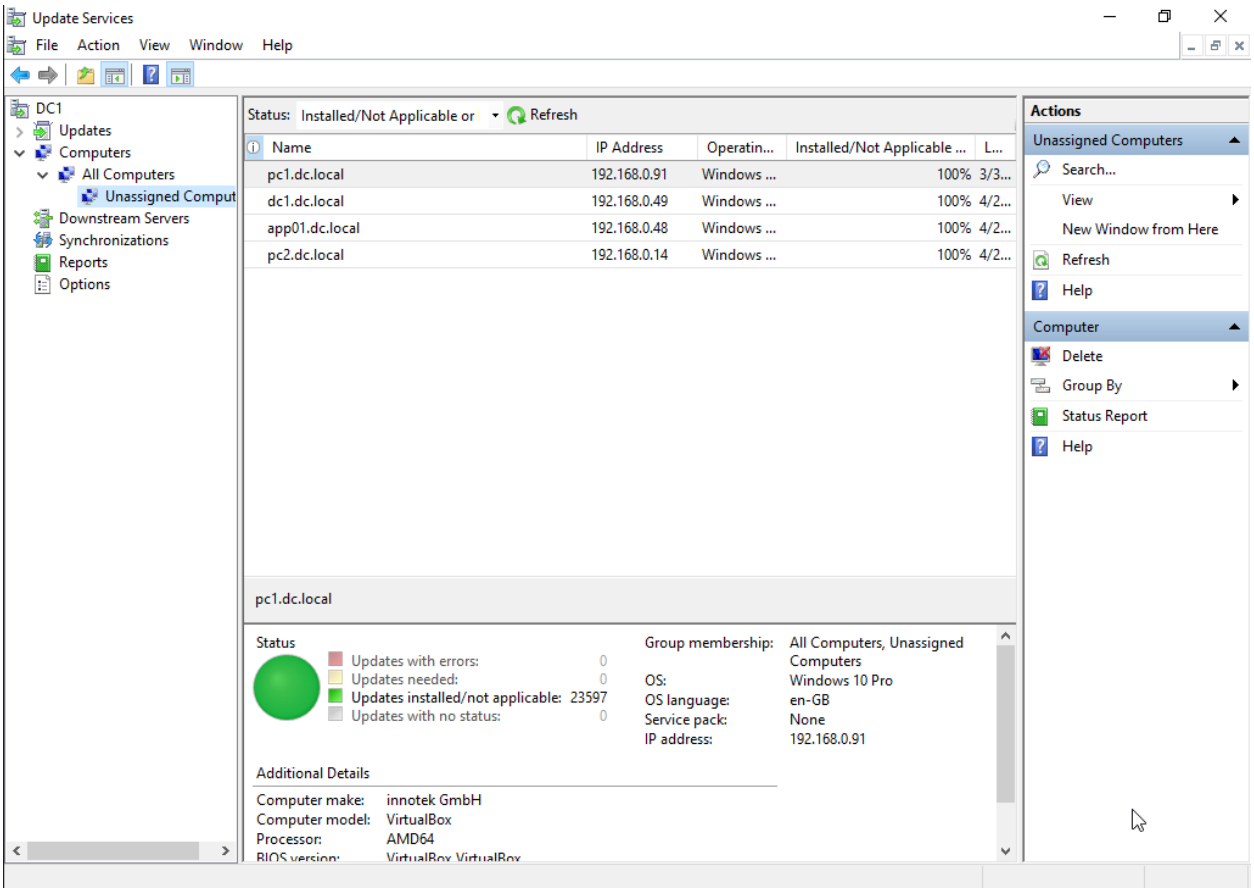
Tasks Completed

- Installed WSUS role
- Configured WSUS server
- Configured GPO:

http://192.168.0.49:8530

- Enabled automatic updates
- Configured client-side targeting





Issues & Troubleshooting

Issue: Clients Not Receiving Updates

Symptoms:

- WSUS not detecting machines

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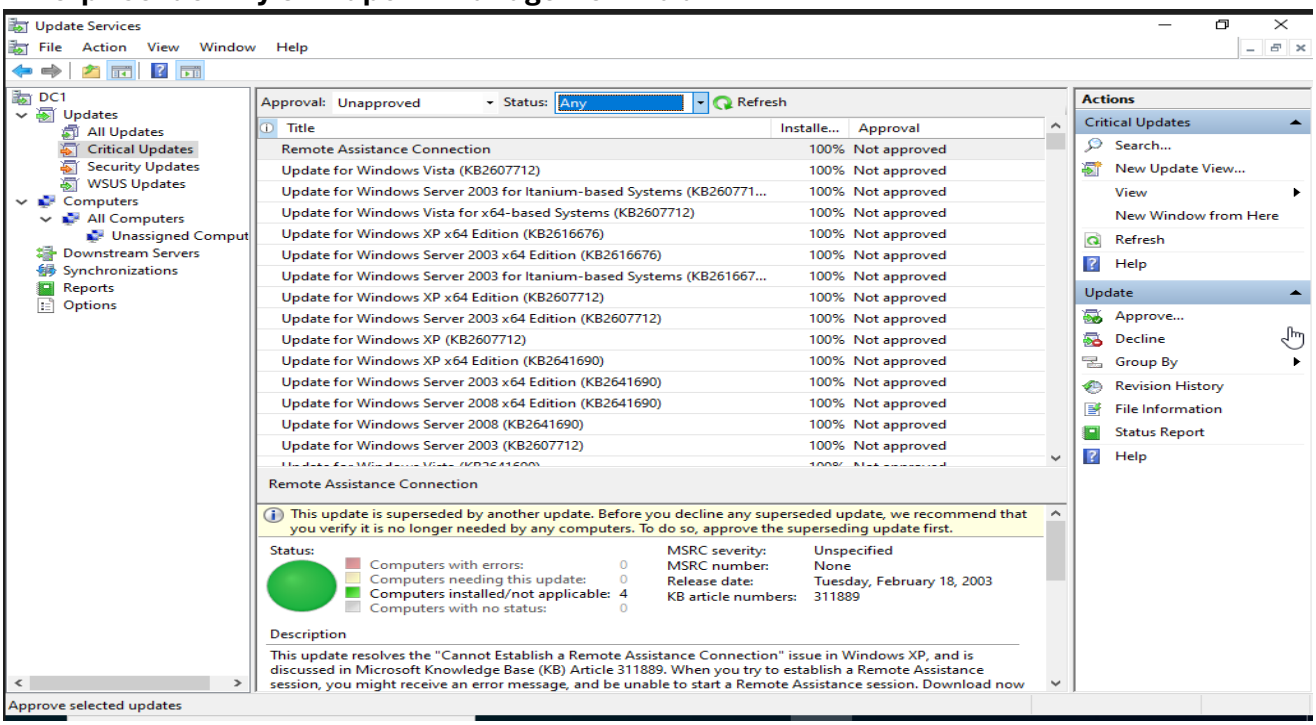

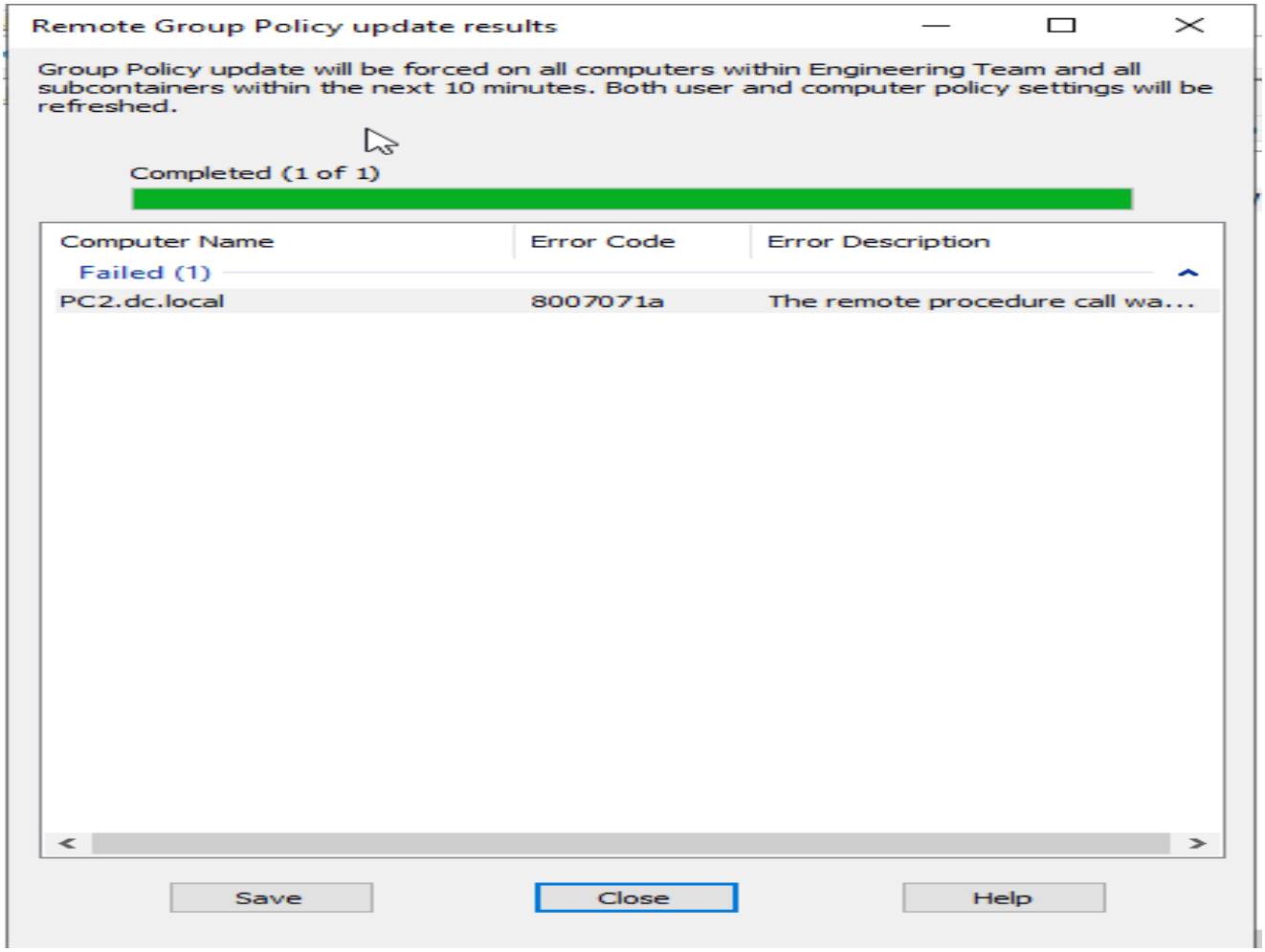

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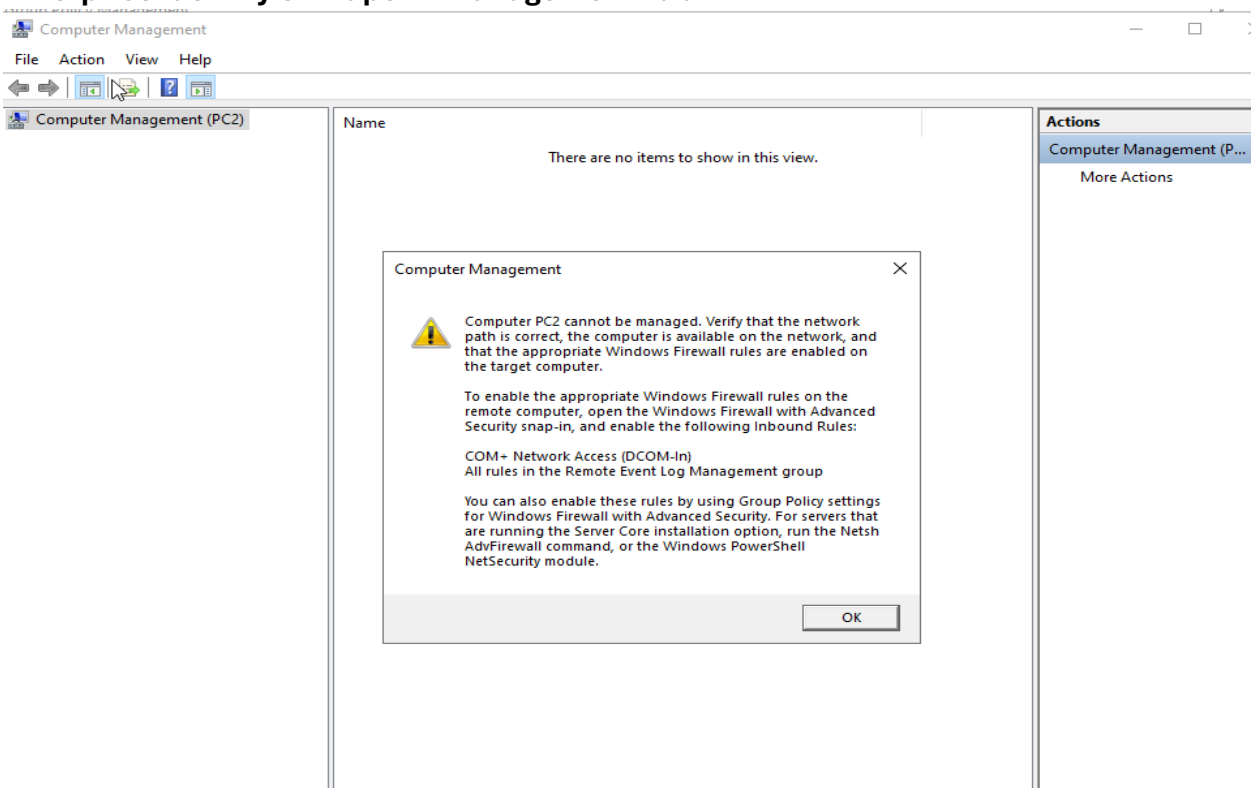

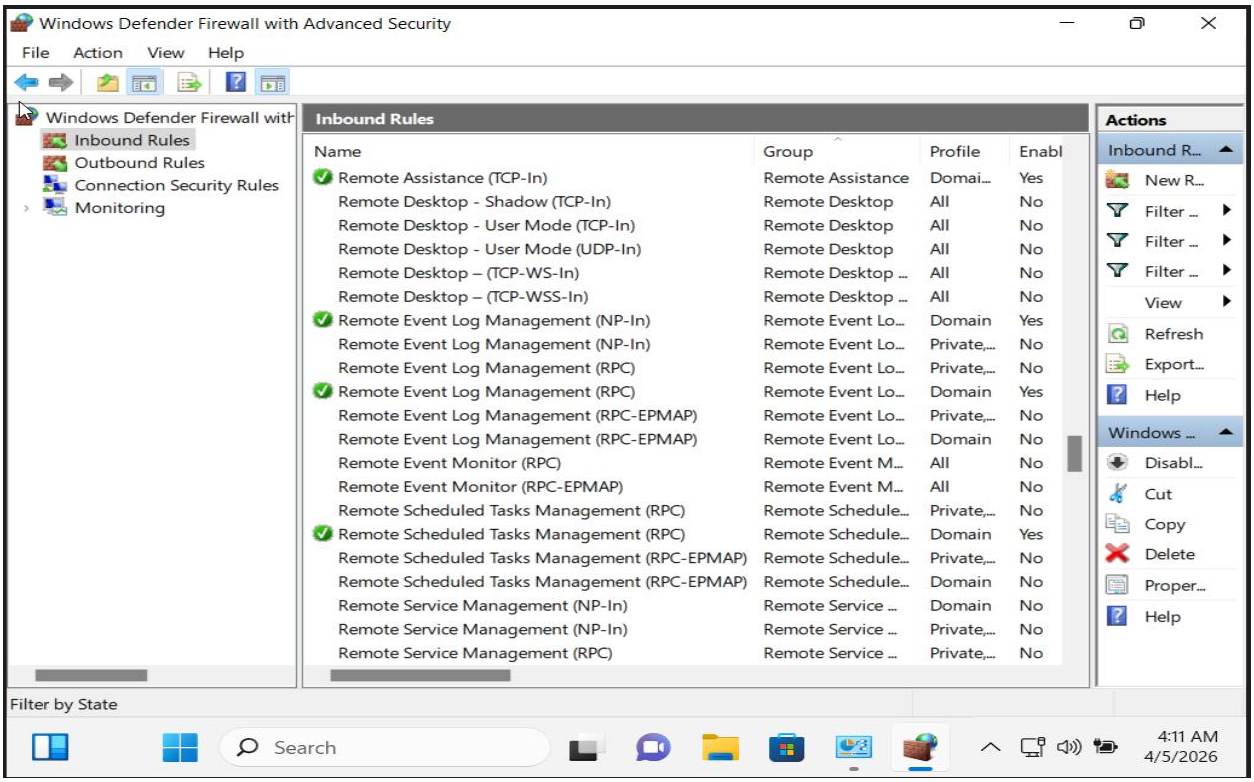

Troubleshooting Steps:

- Verified GPO settings
- Forced update detection
- Restarted client

Resolution:

- Clients successfully connected to WSUS

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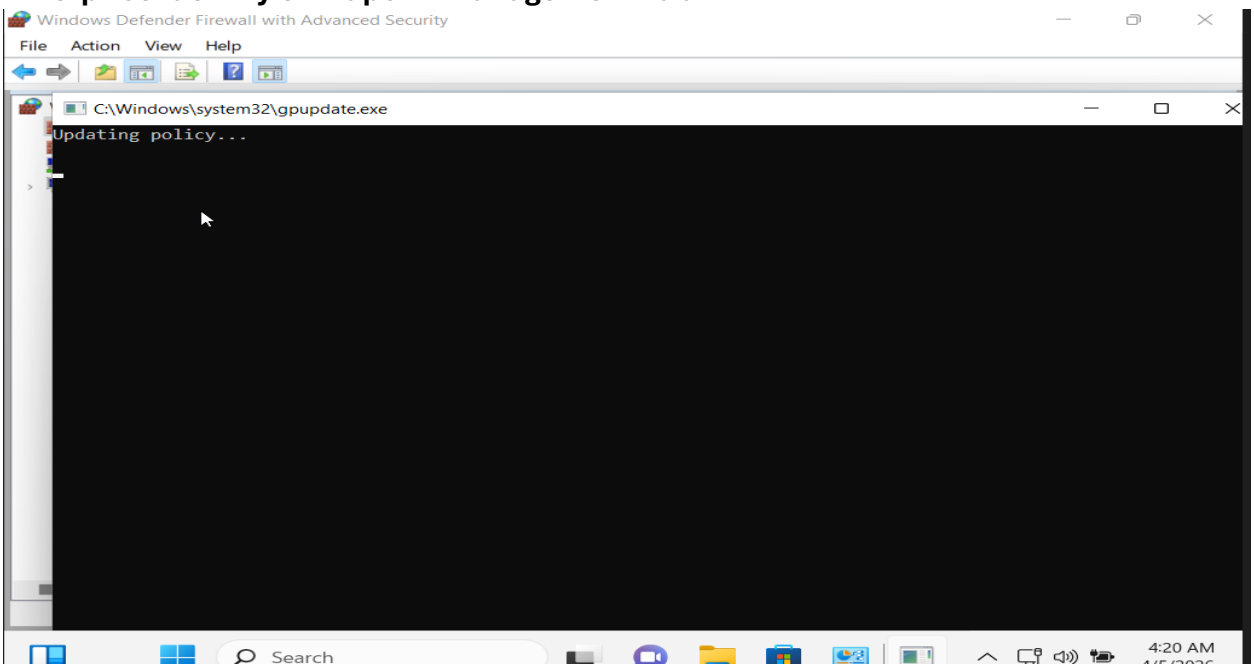


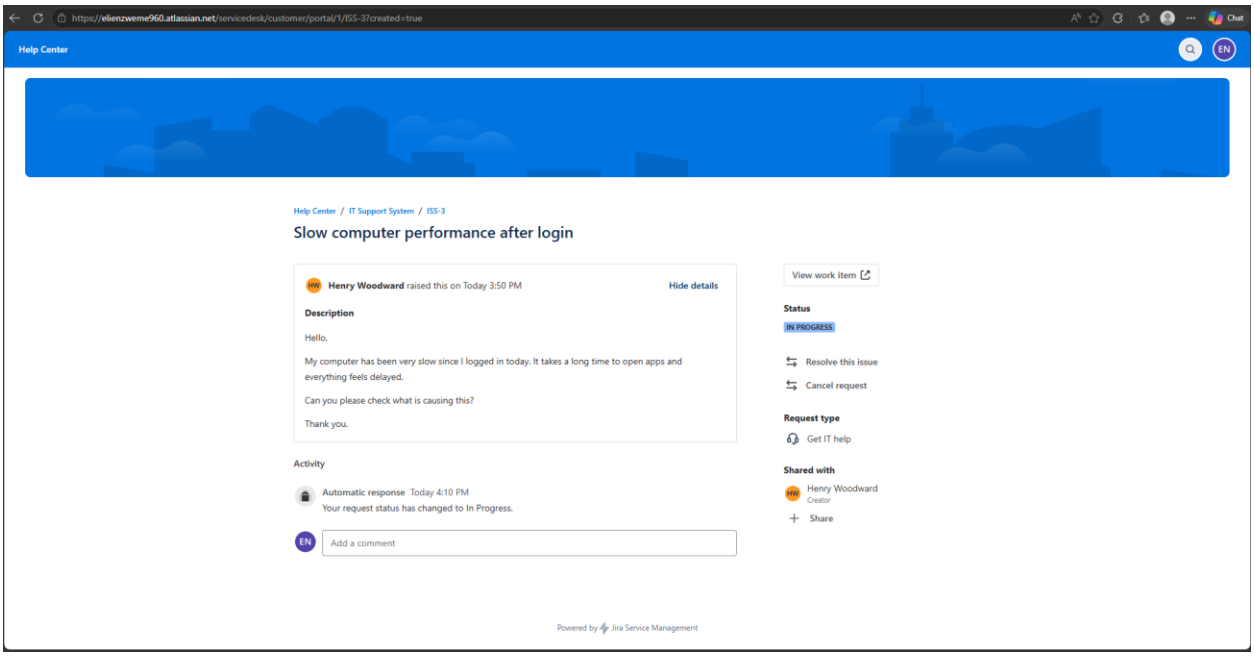
8. Major Incident — Slow Computer Performance

Ticket Summary

Issue:

User reported slow system performance after login:

- Delayed application response
- System sluggishness



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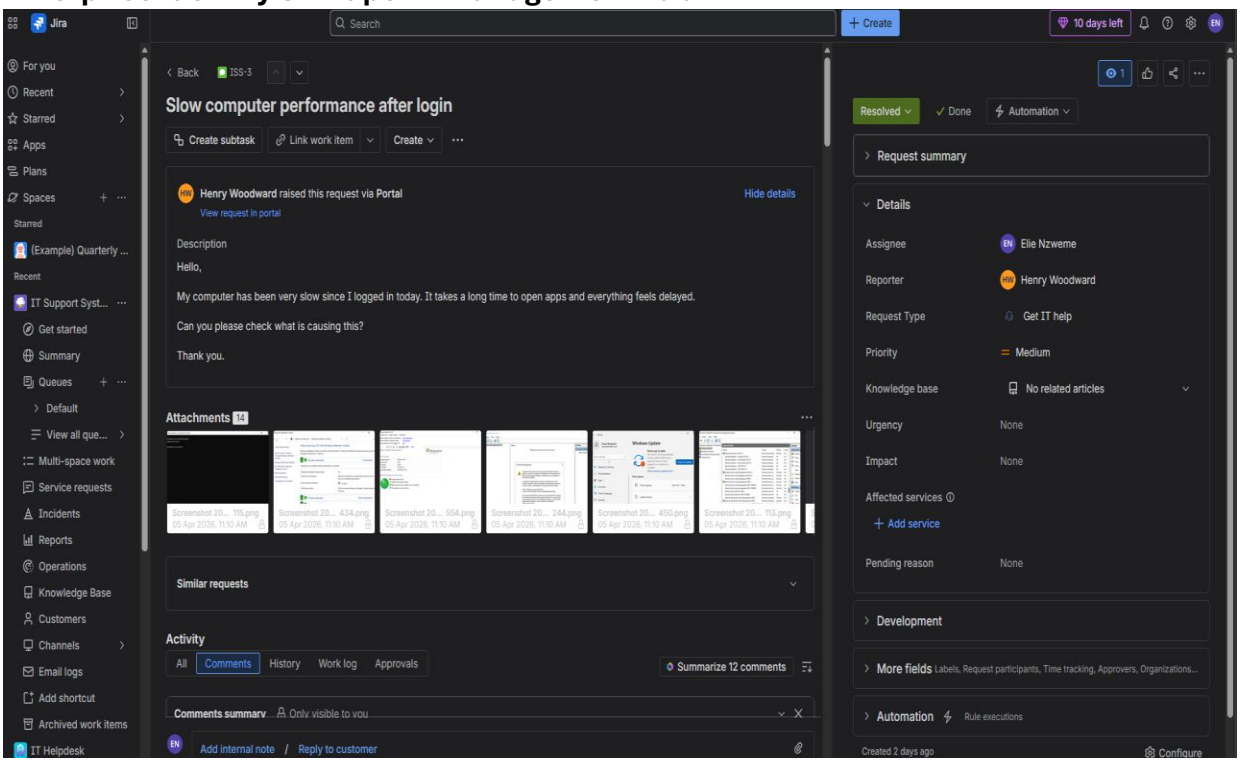
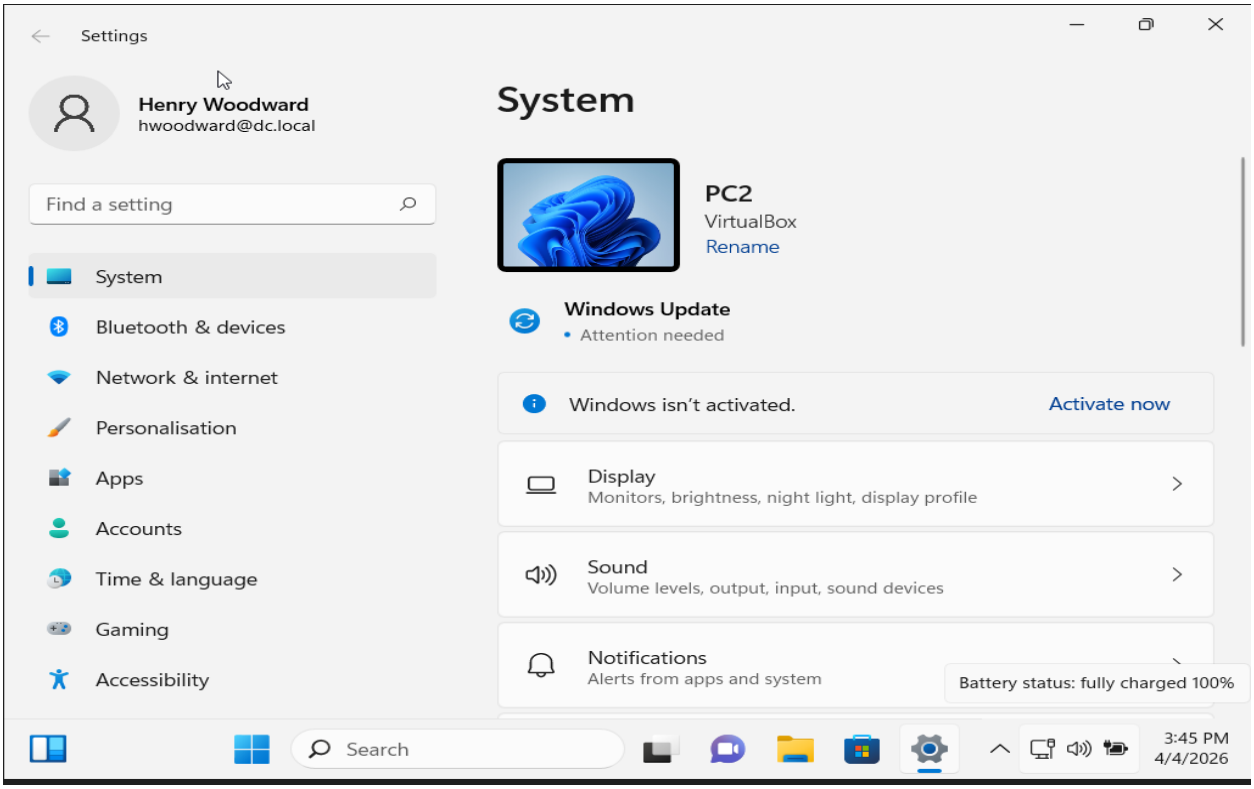


Investigation & Troubleshooting

1. Initial Assessment

- Verified system performance after login
- Checked Windows Update status
- Observed system responsiveness



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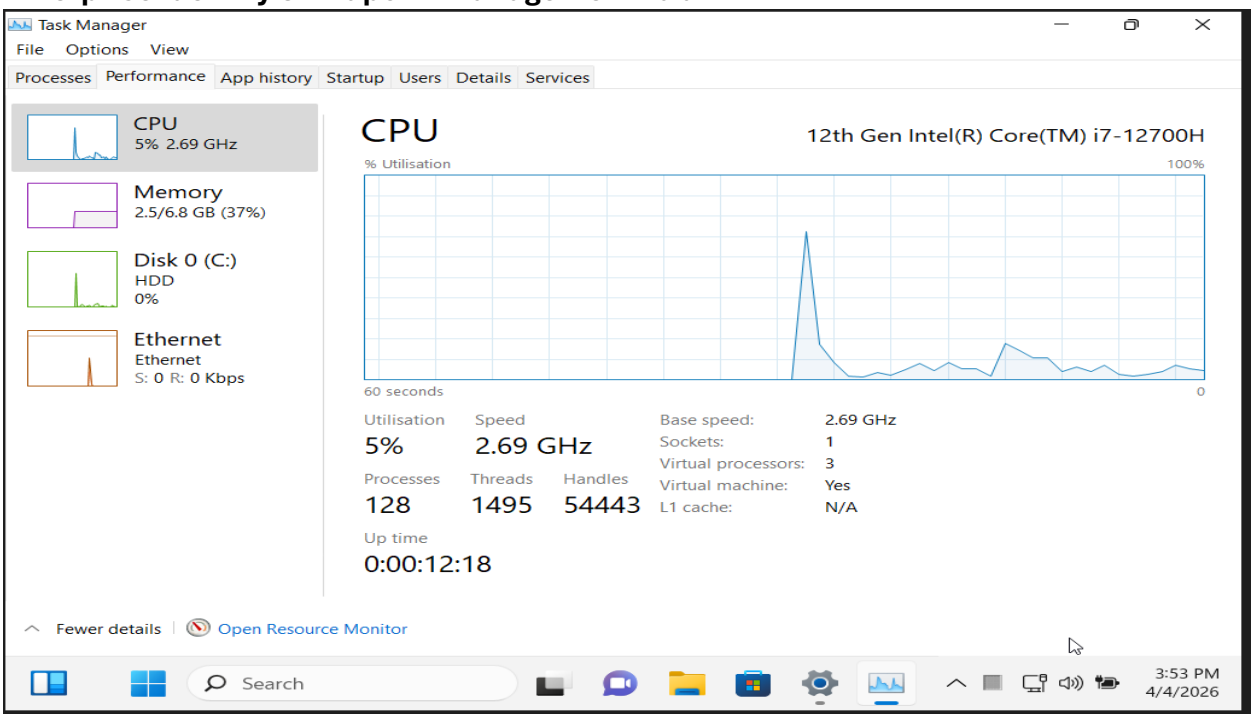
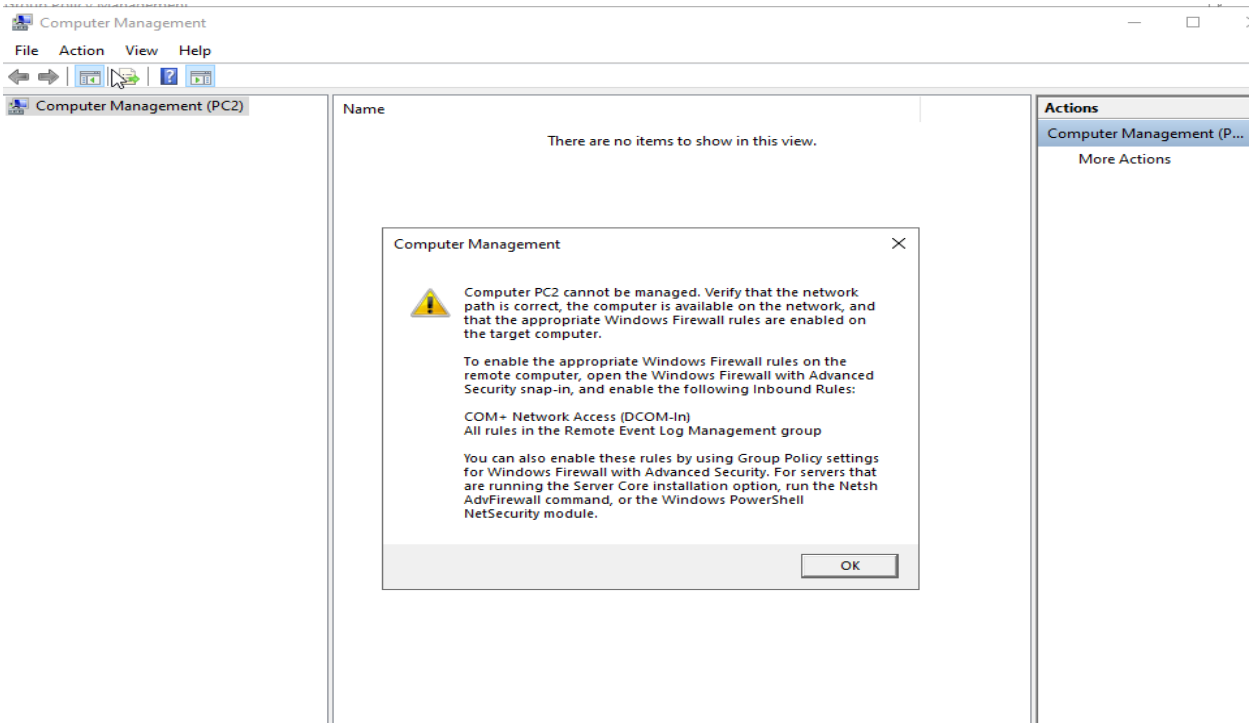


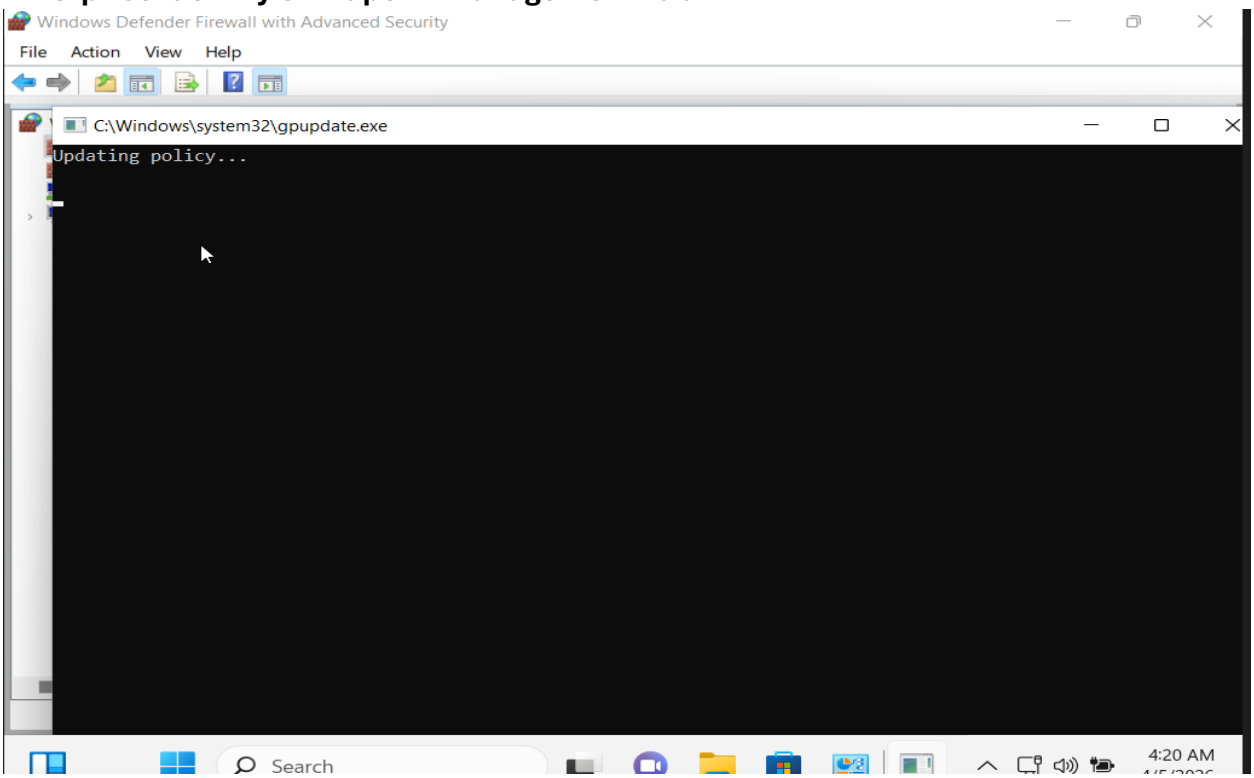
2. Group Policy Verification

- Identified GPO not applying correctly
- Executed:

gpupdate /force

- Observed inconsistent policy behavior





3. Administrative Access Issue

Problem:

- Unable to elevate privileges (UAC/admin issue)

Actions:

- Verified domain admin credentials
- Tested administrative access

Result:

- Elevation worked after system/policy fixes

4. Network & Firewall Troubleshooting

Problem:

- RPC errors during remote management

Root Cause Identified:

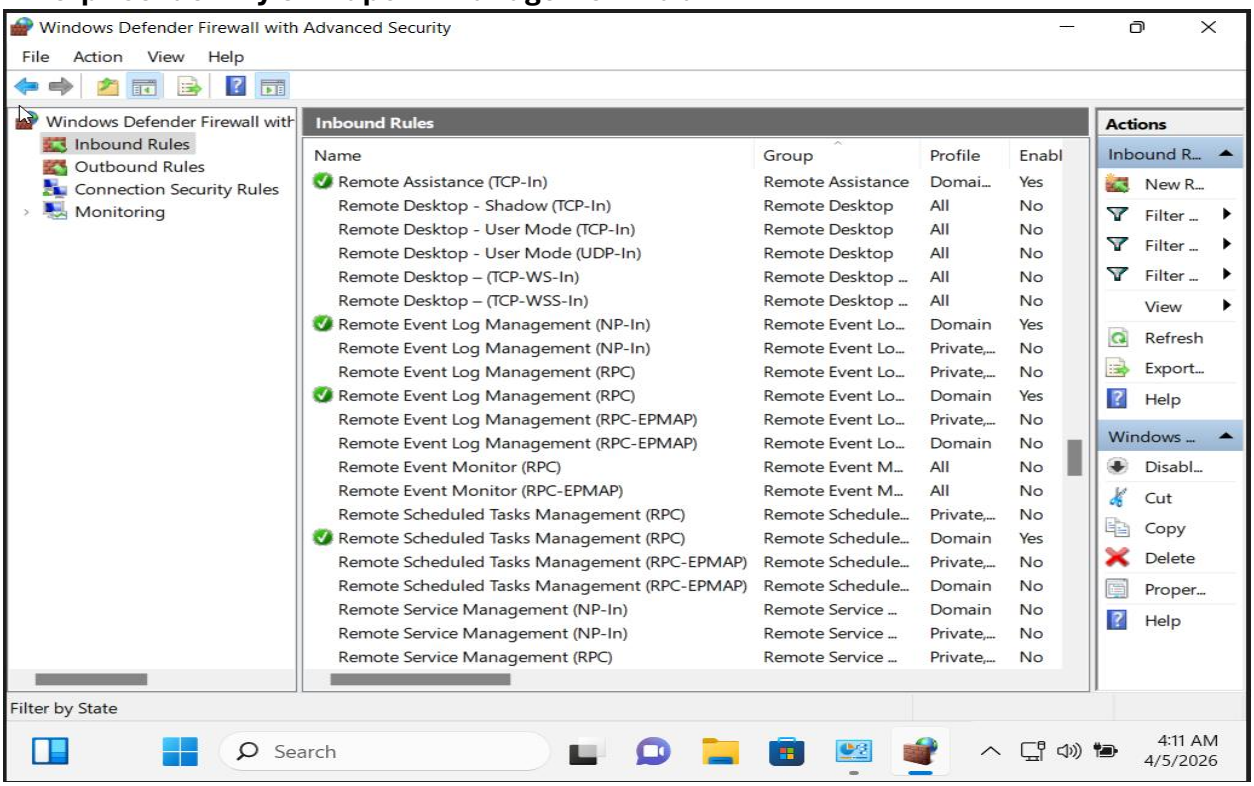
- Firewall blocking required communication

Actions Taken:

Enabled inbound firewall rules:

- Remote Event Log Management (RPC)
- Remote Service Management
- File and Printer Sharing (SMB)

Enterprise Identity & Endpoint Management Lab



5. Group Policy Reapplication

- Ran:

gpupdate /force

- Restarted system

Result:

- GPO successfully applied

6. WSUS Validation

Actions:

- Verified WSUS configuration
- Triggered update detection:

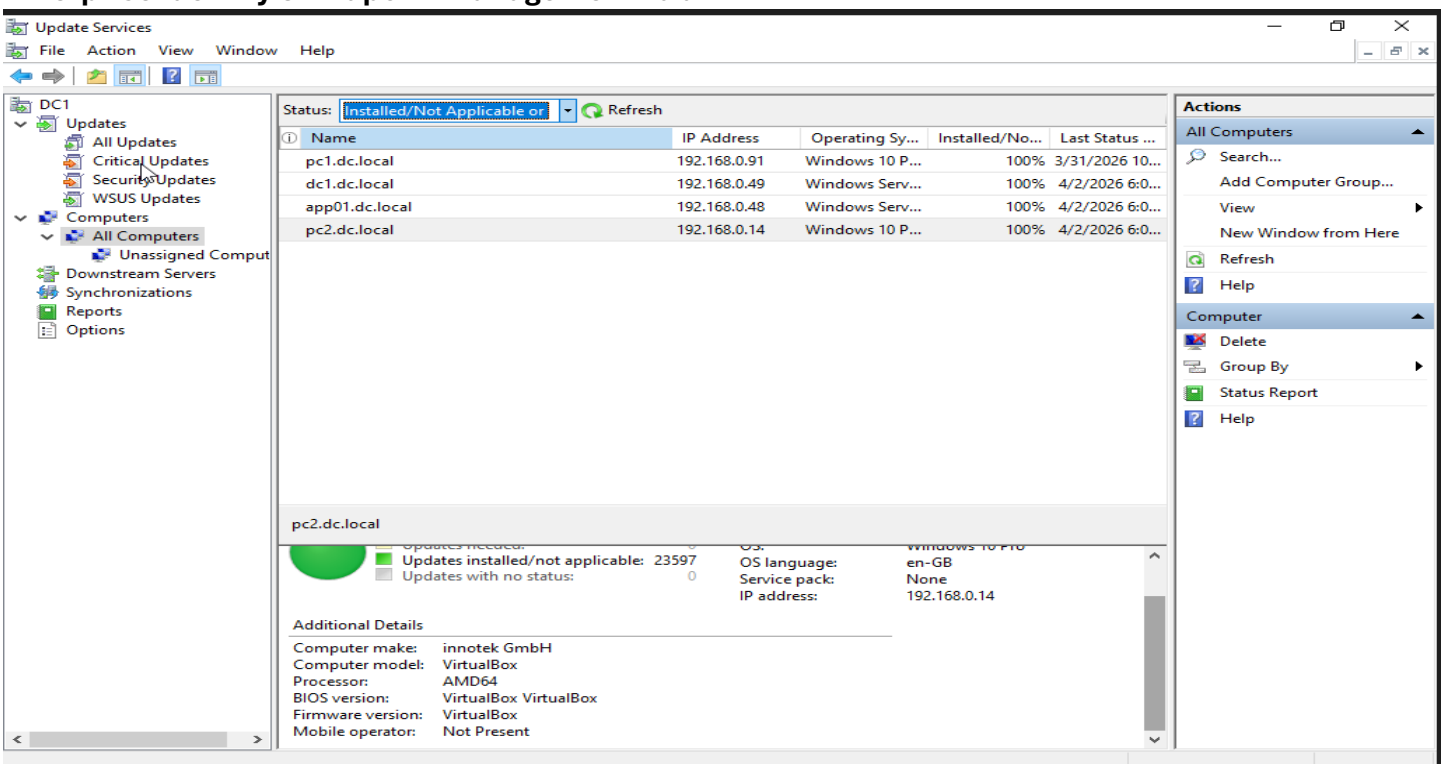
usoclient StartScan

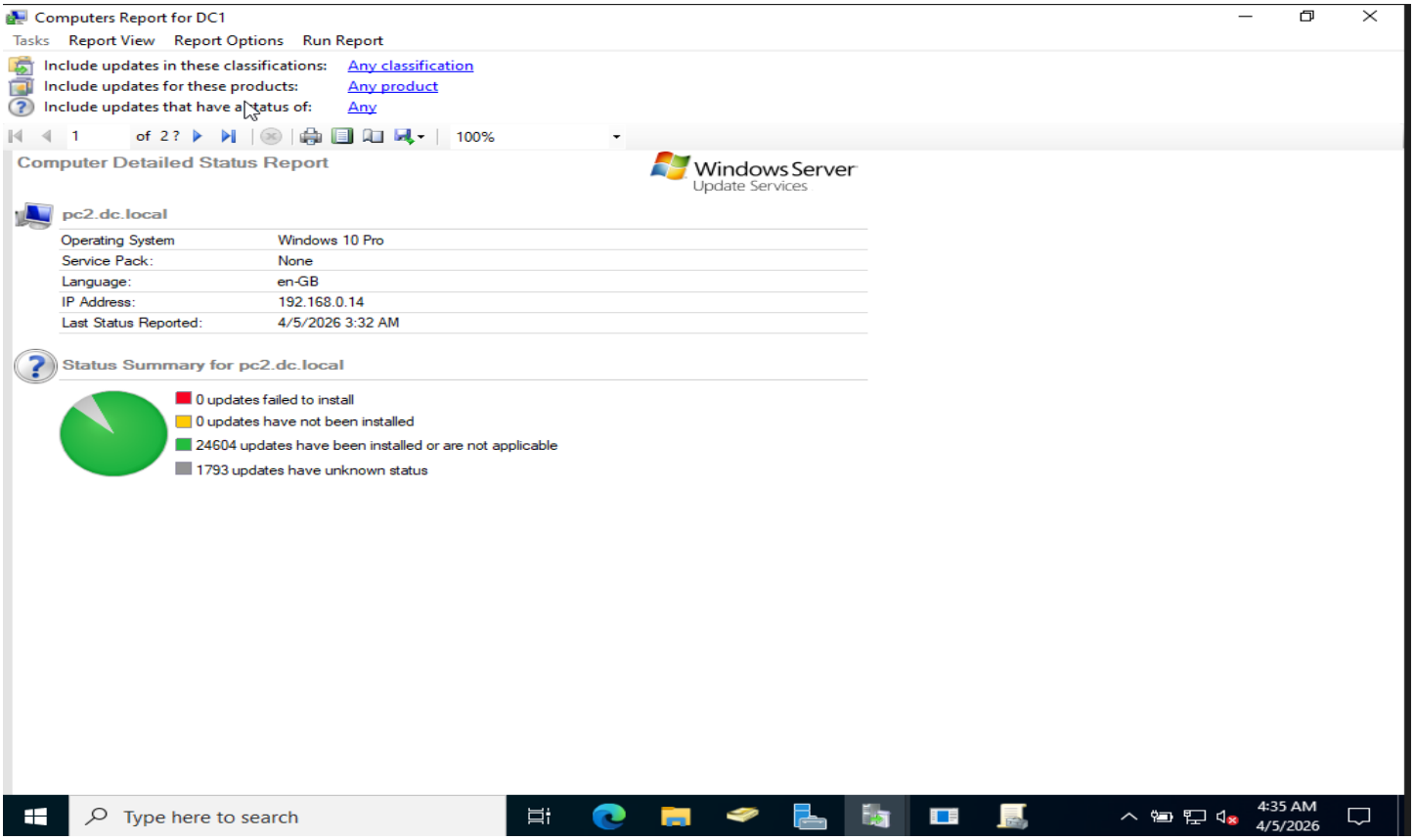
wuauclt /reportnow

Result:

- Client successfully reporting to WSUS

Enterprise Identity & Endpoint Management Lab





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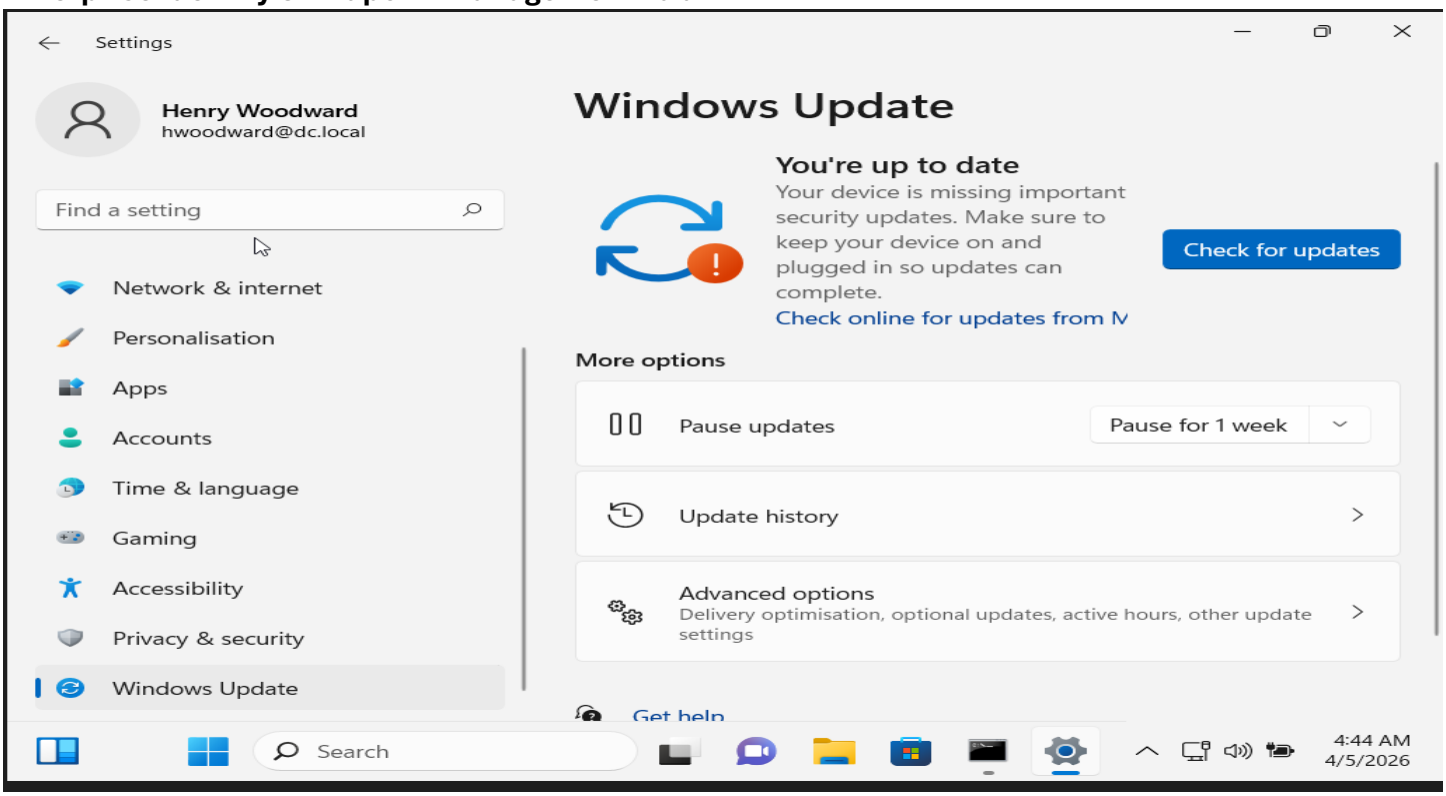

Root Cause

Firewall was blocking RPC communication, which:

- Prevented Group Policy from applying
- Blocked WSUS update policies
- Caused system performance issues

Resolution

- Enabled required firewall rules
- Forced Group Policy update
- Restarted system
- Verified WSUS communication

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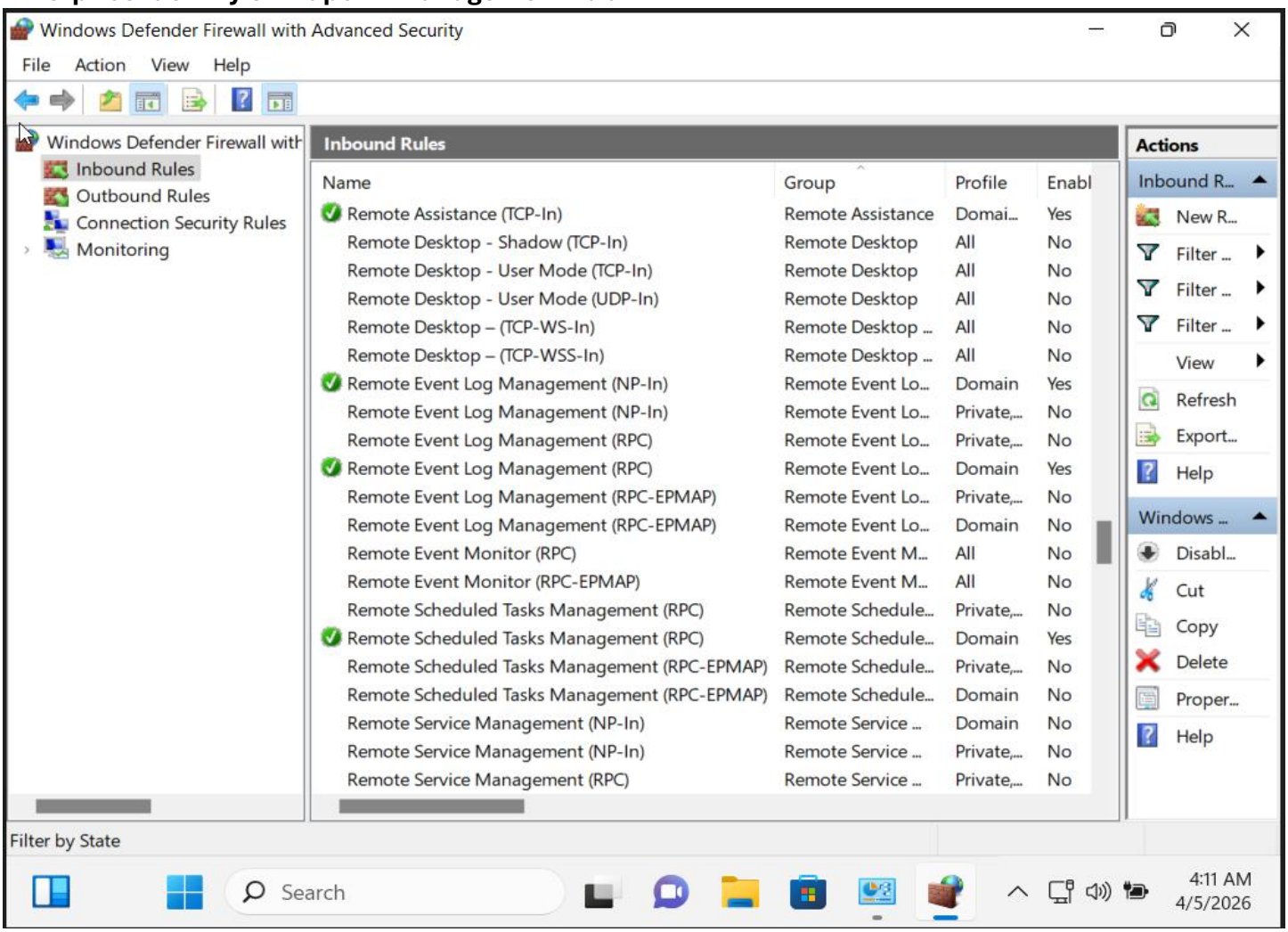
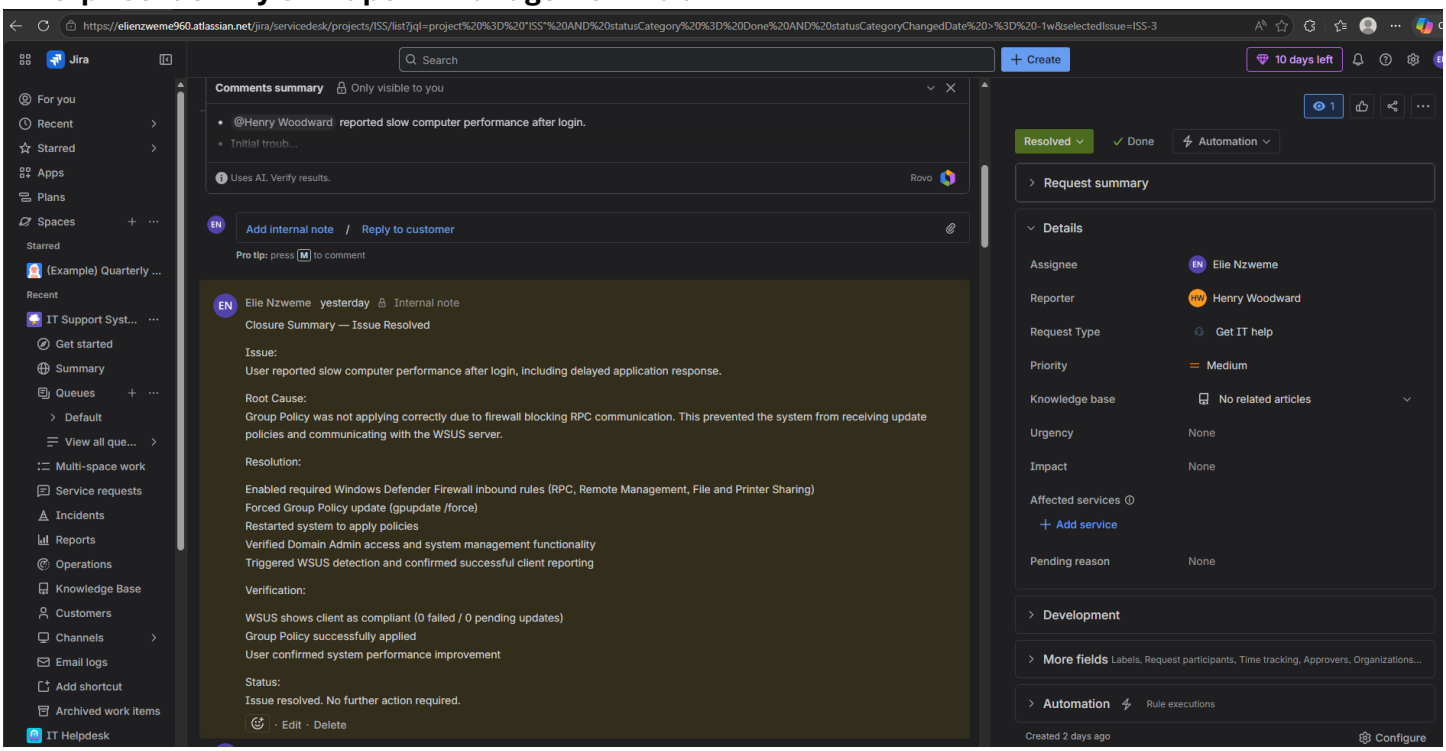


Final Result

- Policies applied successfully
- WSUS fully operational
- System performance restored
- No remaining issues

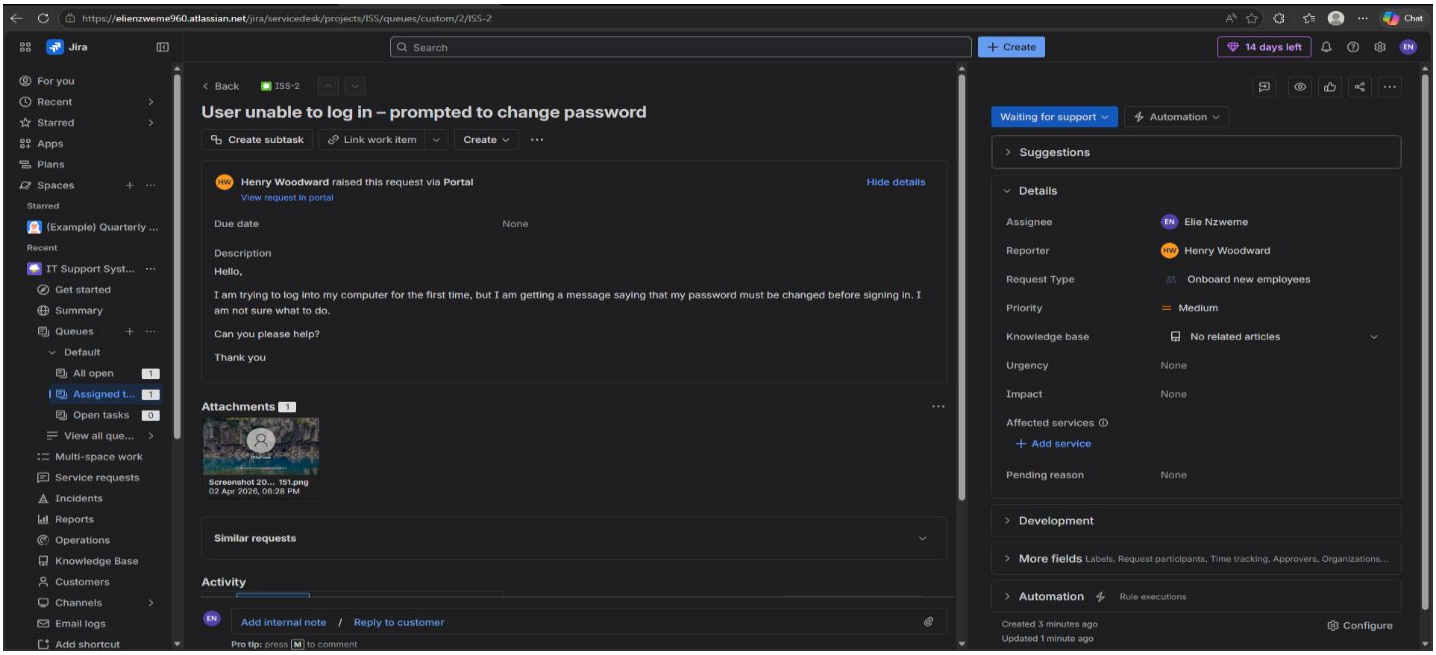


9. Help Desk Tickets (Jira)

Ticket 1: First-Time Login

Issue:

User unable to log in due to password change requirement



Resolution:

- Guided user through password reset
- Verified login

Ticket 2: Shared Folder Access

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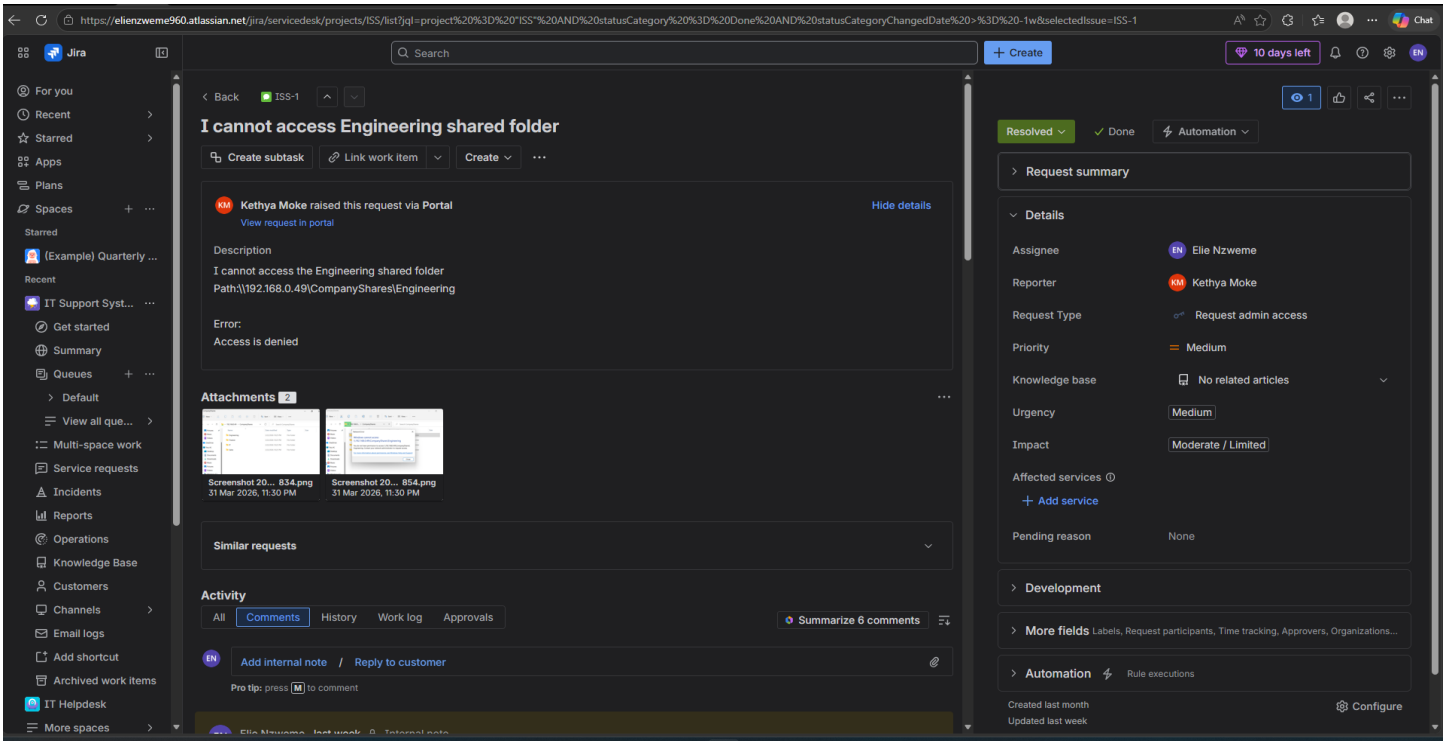
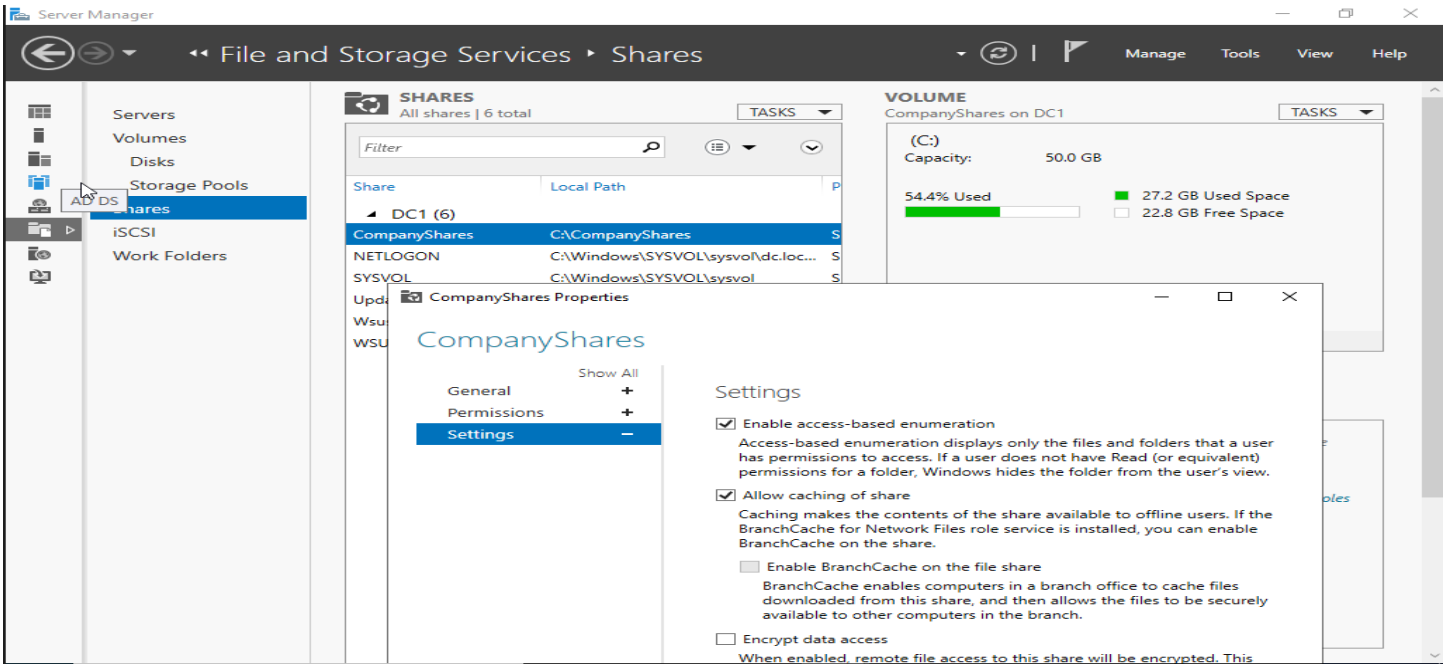
Issue:

User unable to access department folder



Resolution:

- Updated NTFS permissions
- Enabled ABE
- Verified access



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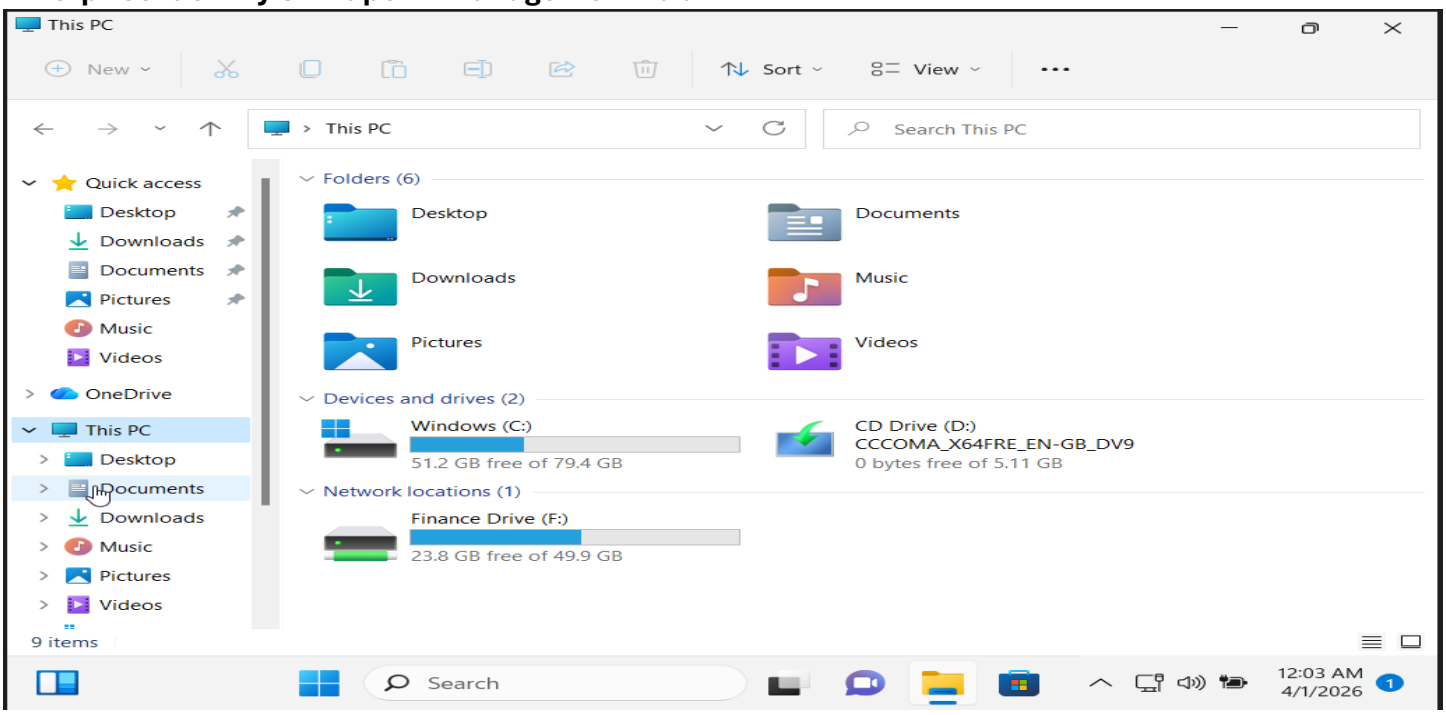

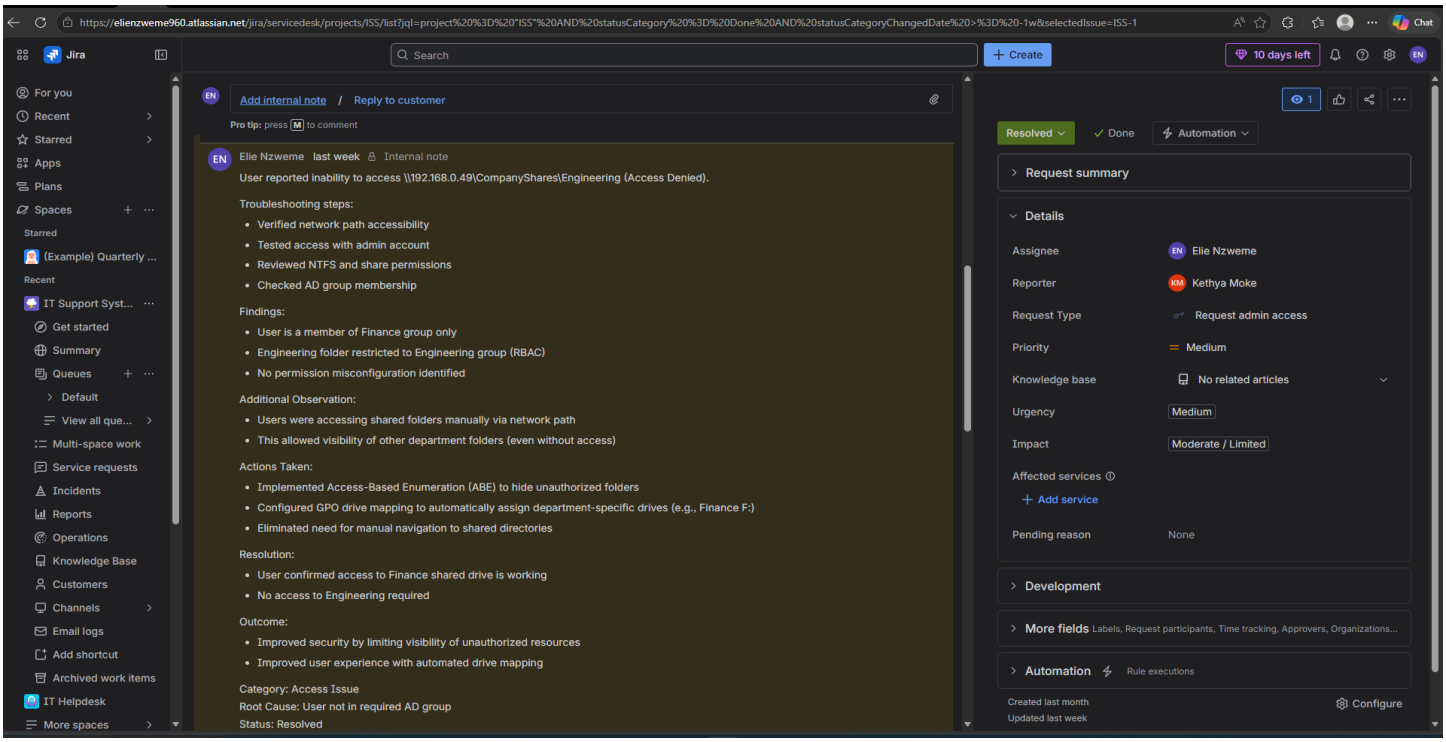

Ticket 3: Slow Computer

Issue:

System is slow after login

Root Cause:

Firewall blocking GPO + WSUS

Resolution:

- Enabled firewall rules
- Forced GPO

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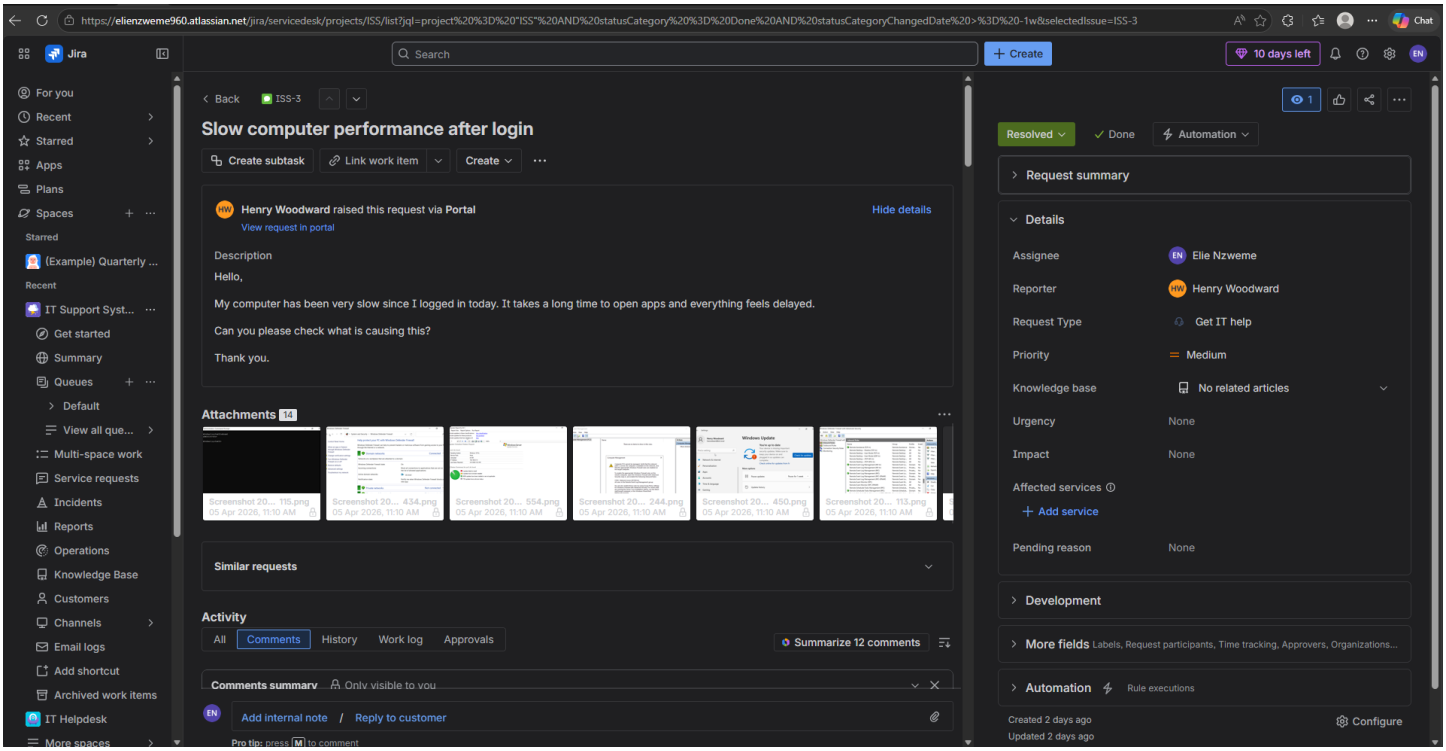
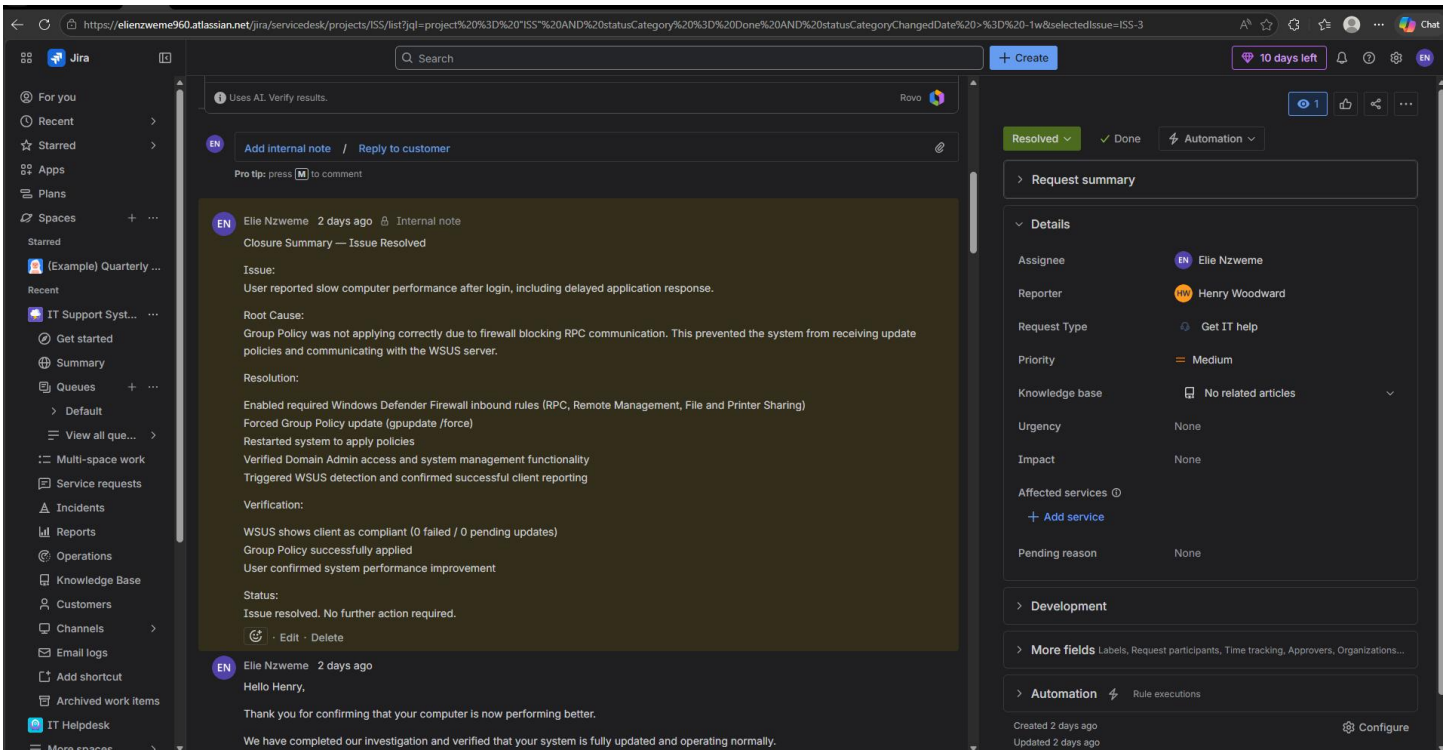
- Verified WSUS
- Restarted system





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10. Troubleshooting Methodology

Throughout the project, I applied:

- Root Cause Analysis
- Step-by-step troubleshooting
- Validation after each fix
- Real-time testing
- Documentation in Jira

11. Skills Demonstrated

- Active Directory Administration
- DNS & Networking Troubleshooting
- Domain Join & Authentication
- Group Policy (GPO)
- NTFS Permissions & ABE
- WSUS Patch Management
- Firewall & RPC Troubleshooting
- Jira Service Management
- System Performance Analysis